Highlights

Does masked-likelihood token gating help image-direction verification under image corruption? A preregistered test in UniDisc

Shlok Kumar Goenka, [AUTHOR: supervising professor, full name]

- Gains or losses above 5 points from token gating excluded under 3 corruptions
- Preregistered test in one model (UniDisc 1.4B), one severity per corruption
- Measures a clean false-alarm share of 0.046 at a target of 0.05
- Finds the gate's likelihood score near chance on JPEG and noise token changes

Does masked-likelihood token gating help image-direction verification under image corruption? A preregistered test in UniDisc

Shlok Kumar Goenka[a], [AUTHOR: supervising professor, full name][a]

[a][AUTHOR: school or department], Vellore Institute of Technology, Vellore, [AUTHOR: postcode], Tamil Nadu, India

Abstract

Unified masked diffusion models treat every visible token as certain, although real inputs can be damaged. We ask whether withholding the image tokens that a pretrained model finds unlikely changes a caption-verification decision. Each image token is scored by its masked likelihood with the caption withheld, a threshold from conformal risk control on clean pairs bounds the expected flagged share of a clean pair's tokens at 0.05, and the gate (GATE-M) drops flagged tokens from an additive evidence sum. Test, sample size and outcome wording were preregistered (exact McNemar, Holm over three corruptions). In the preregistered test no improvement was detected under JPEG quality 10, Gaussian noise with $\sigma = 0.1$ and a 128-pixel splice; for each tested corruption the 90 % CI of the net accuracy change lay inside ± 0.05 (-0.002, -0.006 to 0.000; 0.000, 0.000 to 0.000; -0.004, -0.010 to 0.002; $n = 500$ each), so an improvement larger than 0.05 is excluded and a loss larger than 0.05 is excluded. This is not a finding that GATE-M has no effect. The scope is one pretrained unified masked diffusion model (UniDisc, 1.4B parameters) on COCO image-caption pairs with a random other caption as the negative. On held-out clean pairs the flagged share was 0.046 (95 % CI 0.044 to 0.047): no evidence of violation. A secondary detection family found the gate's score near chance on JPEG and noise (area under the ROC curve 0.561 and 0.503). **[RESULT: one sentence of at most 13 words giving the status of re-infilling (S2) and two-way fusion (B), each with its family label, as run or not run; no causal clause.]**

*[AUTHOR: corresponding author, postal address and published e-mail]

Keywords: Masked discrete diffusion, Conformal risk control, Input reliability, False-alarm control, Vision-language tokens, Fault detection

1. Introduction

Input-token gating asks which of the observed tokens given to a model should be withheld before the model reasons from them. The question arises for any model that reads a sequence of discrete tokens as evidence, and it is sharp for masked discrete diffusion models. A masked diffusion model is trained to predict any masked token from any visible subset of the sequence, and the mask token means "unknown" [1, 2, 3]. Every token that is visible is therefore treated as certain. This is acceptable for a clean training sequence. It is a weakness when the input comes from a pipeline that compresses, adds noise to, crops, splices or mislabels the data, because a single wrong visible token is used with the same confidence as a correct one. Unified models extend the setting to two or more modalities in one token sequence [4, 5, 6]; multimodal diffusion language models such as LaViDa [7] add a vision encoder to a diffusion language model. In such a model a damaged image token can change what the model concludes about a caption, and the reverse.

Why this is a fusion problem.. Each image token and each caption token is a source that reports a value, and the model combines them. The paired decision studied here, whether a caption describes an image, sums per-token log-likelihood ratios over the 1 024 image tokens of one image. That is the rule used to fuse local decisions in distributed detection, in which each source is weighted by its reliability before a global threshold is applied [8, 9]; it is optimal for sources that are independent given the hypothesis, which image tokens are not, so here it is an approximation (Section 6.3). Imperfect and spurious data are a first-order difficulty of multisensor fusion [10], as is falsified data at a fusion centre [11], a survey in this journal organises multimodal fusion on low-quality data into noisy, incomplete, imbalanced and quality-varying settings [12], and evidential fusion methods weigh each modality by an estimated importance [13] or discount it by a learned reliability [14]. Classical practice also checks each source for a fault and isolates the faulty ones before fusing [15, 16], and sets the detection threshold from clean data so that the false-alarm rate stays at a chosen value, as in constant false-alarm rate (CFAR) detection [17, 18]. Here the image tokens are the sources, the masked denoiser is the model of what a healthy source looks like, and the gate

is the fault detector that runs before the evidence is summed. We call this input-level source validation: the gate acts on the sources before the evidence sum, not on the output decision. The paper proposes no new fusion rule; it specifies a gate, a threshold with a stated guarantee, and a preregistered measurement of whether withholding the flagged tokens changes the decision.

A measured failure case.. The gate helps only if the model's own masked likelihood separates damaged tokens from intact ones, and a detection measurement made before the main test shows that it does so for one kind of damage and not for another. Tokens replaced by uniform random codes are separated almost perfectly (mean per-item AUROC 0.961 for a replaced patch and 0.955 for swapped codes), whereas the score is close to chance on JPEG compression (0.561) and Gaussian noise (0.503), where the changed codes depend on the image content. This is a measurement of the score in a secondary family (S-D, Section 5.7); it does not enter the headline test and is not used to interpret it.

What exists and what is missing.. Many recent methods remask tokens in a masked diffusion model. Most act on tokens that the sampler itself produced: they remask low-confidence commitments, learn a quality score, or perturb the context to find brittle tokens [19, 20, 21, 22, 23, 24, 25]. In multimodal masked diffusion models the same idea has been applied to the model's own generations, with cross-modal confidence as the trigger [26, 27]. Work on corrupted *input* tokens is thinner. Trust but Verify [28] detects corrupted conditioning tokens in a text-only masked diffusion model, re-noises them to the mask state and refills them; it needs a finetune, uses a hand-set threshold, and has no second modality. Conformal risk control has been used with masking-derived scores to flag erroneous caption words given an image, with a CLIP-based score and a threshold calibrated on labelled foil data [29]. In this journal, Zhou and Lu [30] describe, in their abstract, a gate that keeps a default vision-language prediction unless a challenger's Dempster–Shafer evidence score clears a calibrated threshold, with Clopper–Pearson bounds used to set the operating point; its unit is a whole prediction rather than an input token. **[AUTHOR: verify against full text via VIT e-gateway]** The authors of UniDisc show repair of corrupted image-caption pairs by random masking and best-of-n re-infilling in one qualitative figure [4]. Section 2 gives the precise differences. Three things are missing from this picture. No measurement separates a gate that finds damaged tokens from a mask that merely has the

same shape. No threshold for input tokens carries a stated bound on the false-alarm proportion on clean inputs. And no comparison shows whether withholding flagged tokens helps a cross-modal decision more than cheap non-generative detectors or a pixel denoiser do.

Contributions, stated as a measurement.. This paper does not claim to be the first to detect or remask corrupted input tokens, the first to remask in a unified model, the first risk-controlled threshold for vision-language errors, or the first masked-likelihood or null-normalised generative score. The gate is the object of measurement here, not an asset whose merit is assumed. Each numbered item corresponds to a result that the preregistered analysis delivered whatever it found.

1. *The measured object* (Section 3). We specify an input-level source-validation gate for a pretrained unified masked diffusion model: each image token is scored by the model's own masked likelihood with the caption withheld, and the flag threshold is set by conformal risk control on clean pairs so that the expected flagged share of a clean pair's tokens is at most α.
2. *A closed-form condition* (Section 3.7). Under an independent-token approximation, we give the condition on the false-alarm and hit rates under which discarding flagged tokens raises the signal-to-noise ratio of an additive evidence sum; it is an approximation because image tokens are dependent, and it is not linked to any result of the primary test.
3. *A preregistered test* (Sections 4 and 5.1). An exact McNemar test with Holm correction over three realistic corruptions, with the sample size fixed by a blind headroom rule, of whether withholding the flagged tokens changes image-direction verification accuracy, with controls for mask geometry, texture and pixel denoising. The preregistered test excludes an improvement or a loss of more than five percentage points in image-direction verification accuracy from gating under JPEG quality 10, Gaussian noise with $\sigma = 0.10$ and a 128-pixel splice.
4. *Token detection, measured* (Section 5.7). At the token level the gate's score separates code-level random replacements (AUROC 0.961 for a patch, 0.955 for swapped codes) but is near chance on JPEG and Gaussian noise (0.561 and 0.503). On noise it is below a one-pass score without finetuning (0.607; the marginal 95 % intervals, 0.493–0.513 and 0.597–0.618, do not overlap; no paired test was run), and on JPEG

the two overlap (0.561, 0.547–0.575, against 0.581, 0.569–0.593). In the preregistered comparison it is below a texture score whose sign was chosen on the same items (0.756 and 0.768), so the preregistered falsifier for the score was met.

5. *Calibration, checked* (Section 5.5). We check the clean false-alarm statement on held-out clean test pairs. At $\alpha = 0.05$ the realised flagged share was 0.0456 (95 % interval 0.0442 to 0.0470), with no evidence of violation.
6. **[RESULT (A4.2.4 item 6): *Secondary families* (Sections 5.10 and 5.11): name only those of re-infilling (S2) and two-way fusion on hard negatives (B) that ran, with their status; delete this item if neither ran.]**

The scope is one pretrained 1.4B model, one image-caption dataset, three realistic image corruptions in the primary family and two sanity corruptions in a separate positive control (PS). The primary endpoint is the image-direction statistic, which has headroom on clean pairs; the text direction is near ceiling there, and Section 4.8 states what this choice limits. Caption corruption is not tested.

[AUTHOR: Figure 1 (motivation, page 1-2 per IF anatomy rule 31) is being prepared separately from committed run arrays; do not draw it with a generative image tool.]

Roadmap.. Section 2 places the work against the closest methods. Section 3 defines the scores, the calibrated gate, the evidence statistic and the algorithms. Section 4 gives the data, folds, corruptions, arms, controls, the families and the reading of the primary test; the preregistration record and the deviations are in Appendix A. Section 5 reports the results family by family, Section 6 discusses them with the limitations, and Section 7 concludes. Appendix B gives the evidence that the model has no time input in the released configuration.

2. Related work

The related work is organised along four axes: which tokens are treated as suspect, how a token is scored, how the threshold is set, and what is done with a flagged token. Table 1 lists the closest works on these axes.

Masked diffusion and unified models.. Absorbing-state discrete diffusion [1, 31] was simplified into masked diffusion language models [2] and scaled [32]. A masked diffusion denoiser models the conditional distribution of a clean token given the unmasked context [3], which is the property the gate score uses. Masked image-token generation [33] and unified text-image models [4, 5, 6] put several modalities in one sequence; multimodal diffusion language models such as LaViDa [7] add a vision encoder to a diffusion language model. UniDisc, the backbone used here, is trained to mask all tokens of a modality with some probability, so that a modality likelihood with the other modality withheld exists [4]. The appendix of that paper demonstrates repair of overlaid and mismatched image-caption pairs by random masking of both modalities, re-unmasking and best-of-n selection by the model's own likelihood, shown as examples without a metric or a per-token score. That recipe is the closest prior procedure for re-infilling and is treated as a baseline in Section 4.

Remasking and self-correction of generated tokens.. Training-free remasking acts on the sampler's own commitments [19, 23, 24]. Learned variants add a per-token quality score or a correction head [20, 21, 22]. For masked image generation, a learned critic decides which sampled image tokens to resample [25]. In multimodal masked diffusion models, self-correction re-evaluates committed tokens with the token remasked and remasks the lowest-scoring ones [26]. A coupled sampler for concurrent text and image generation retracts commitments when cross-modal evidence turns against them [27]. All of these act on tokens that the model produced. The tokens studied here are given evidence whose provenance is untrusted, which raises a question that does not arise for self-correction: whether a suspect token should be withheld (left masked) or imputed (re-infilled). The loop of detecting, remasking and refilling is not claimed as new, and the cross-modal remasking trigger belongs to the works above.

Corrupted input and conditioning tokens.. Trust but Verify [28] is the nearest work. It studies corrupted context tokens in a masked diffusion language model with 170M parameters, finetuned with substituted context tokens. Its suspicion score is the model's probability of the observed token at its unmasked position, its threshold is annealed by hand within an edit budget derived from an assumed corruption rate, and flagged tokens are re-noised to the mask state and refilled. The paper reports a false-edit rate on clean tokens, and it reports that corruptions that are plausible under the left

context are only partly detected even with the finetune. The present work differs in five ways: no finetuning of the network, a score computed after masking the token, with the other modality withheld, a second modality, a threshold with a stated bound on the clean false-alarm proportion, and an outcome measured on a cross-modal decision rather than on infilling quality. Its observation that detection of plausible corruptions is hard is treated here as an expected difficulty for JPEG and noise, with a stated falsifier (Section 4). That paper also suggests that bidirectional consistency checking is needed for plausible corruptions; the masked score used here is one way to make that check. Because the finetune is what makes the observed-position probability informative in that paper, the baseline named "TbV-score (non-finetuned)" below reproduces its score but not its method.

In vision-language models, uncertainty-guided dropout decoding masks visual tokens that are uncertain under a logit-lens projection, with hand-set drop probabilities and without a calibrated threshold [34]. A self-purification defence masks the visual tokens of highest saliency to purify poisoned training responses before finetuning [35]. Neither evaluates detection of corrupted tokens against known labels.

Masked-likelihood scores, null-normalised scores and calibrated thresholds.. Scoring a token by its masked likelihood has a long history in text: the pseudo-log-likelihood [36], its variant for multi-piece words [37], and unsupervised spelling detection from the probability of the original token under a masked language model [38]. A generative model can be used for image-text matching by normalising the conditional error with the unconditional one [39], a null-normalised score that the pointwise-mutual-information statistics of this paper resemble. None of these is claimed as new. Conformal prediction supplies distribution-free guarantees under exchangeability [40, 41], conformal p-values give false-discovery control for outlier detection [42], and conformal risk control bounds the expected value of a monotone loss [43]. Gomes et al. [29] apply conformal risk control with masking-derived CLIPScore distributions to flag erroneous caption words. Their guarantee is a high-probability upper bound (Hoeffding and Bentkus) calibrated on foil-labelled data, with a CLIP score that changes when words or patches are masked. The calibration here uses clean pairs only, bounds an expectation, nests token losses inside items, and scores tokens with a generative within-modality likelihood, including image tokens. It does not make the guarantee of Gomes et al. the first of its kind, and it is not a stronger one: the two answer different questions.

Source validation before fusion.. The classical background is outside machine learning. Failure detection and isolation [15, 16], CFAR thresholding [17, 18, 44], decision fusion with weighted reliabilities [8], and combination rules for classifiers [45] are the vocabulary the method borrows. Withholding a set of tokens selected by looking at their own values makes the missingness informative [46]; Section 3 states the conditions under which withholding a data-selected set is still the right treatment.

Work in this journal treats unreliable inputs mostly by weighting them. Evidential fusion methods attach an estimated importance [13] or reliability [14] to each modality and combine the modalities with Dempster–Shafer theory; the reliability version discounts each modality before Dempster's rule [14]. Discounting is the soft counterpart of withholding a source, and Section 4.11 states a pre-specified exploratory comparison of the hard gate with it. A survey of multimodal fusion on low-quality data lists noisy and quality-varying inputs among its four settings [12]. Two recent papers in the journal bound the risk of trusting a source at the level of a whole decision. As described in its abstract, Zhou and Lu [30] replace a default vision-language prediction by a challenger's only when a Dempster–Shafer evidence score clears a calibrated threshold, with Clopper–Pearson bounds used to set the operating point, and Wang et al. [47] use a conformal prediction method on a text-and-tabular fusion network to flag unreliable predictions for human review. **[AUTHOR: verify against full text via VIT e-gateway]** Both act on the output of a recogniser or classifier; as far as their abstracts describe, neither scores the individual input tokens of one modality, and neither uses the likelihood of a generative model. The present gate differs in its unit (an input token, before any fusion), its score (the masked likelihood of the token inside its own modality with the other modality withheld) and its guarantee (a conformal-risk-control threshold calibrated on clean pairs, which bounds the expected false-alarm proportion on clean inputs and uses no corruption labels). **[AUTHOR: the comparison with Zhou and Lu rests on its abstract and preview, because the full text was not available; read it (risk definition, labels needed for calibration, any corruption experiment) and upgrade or correct this paragraph before submission.]**

3. Method

This section defines the notation, the scores, the calibrated gate, the evidence statistic and seven algorithms. Each algorithm calls the earlier ones,

and Algorithm 7 composes all of them. The method has three steps. Each input token is *scored* by its masked log-likelihood inside its own modality with the other modality withheld. Tokens whose score is improbable for clean inputs, at a threshold set on clean data, are *gated*. The decision then *fuses* evidence from the tokens that remain (GATE-M, the marginalise arm), or from tokens re-infilled by the model (GATE-I, the impute arm; the SAP calls it I-GATE). Cost is counted in network passes, listed per quantity in Appendix C. One design rule organises the three steps: evidence that is tested must not be used to gate (Section 3.3). The denoiser is used through one conditional only, the probability of a token given a visible subset, called with no time input; Section 4.1 and Appendix B state what is checked about that property for the released checkpoint.

3.1. Setting and notation

A unified masked discrete diffusion model with absorbing-state masking is given a sequence of caption tokens and image tokens. The network p_θ maps a partly masked sequence to a categorical distribution at every masked position. A pair is $x = (t, v)$ with caption tokens $t = (t_1, \ldots, t_n)$ and image tokens $v = (v_1, \ldots, v_m)$. The modalities are $a, b \in \{T, I\}$ with $a \neq b$. Table 2 lists the symbols. All log-likelihoods are in nats, computed as an `fp32` log-softmax of the logits restricted to the vocabulary of the scored modality.

3.2. Masked scores

For a token j of modality a with observed value x_j, four context levels are distinguished. Let $\Gamma_{k(j)}$ be the group of j in a partition Π_K, and let $x^a_{-\Gamma}$ denote the tokens of modality a outside Γ.

$$\begin{aligned} A_j &= \log p_\theta\big(x_j \mid x^a_{-\Gamma_{k(j)}},\, x^b\big), & B_j &= \log p_\theta\big(x_j \mid x^a_{-\Gamma_{k(j)}},\, \varnothing^b\big), \\ C_j &= \log p_\theta\big(x_j \mid \varnothing^a,\, x^b\big), & D_j &= \log p_\theta\big(x_j \mid \varnothing^a,\, \varnothing^b\big). \end{aligned} \tag{1}$$

A_j uses the rest of the own modality and the other modality; B_j uses the own modality only; C_j uses the other modality only; D_j uses nothing and is the position marginal. By subtraction, the total pointwise mutual information splits in two ways,

$$\begin{aligned} A_j - D_j &= \underbrace{(B_j - D_j)}_{q_j\text{: intra-modal}} + \underbrace{(A_j - B_j)}_{\text{cross-modal given own context}} \\ &= \underbrace{(A_j - C_j)}_{\text{intra-modal given } b} + \underbrace{(C_j - D_j)}_{\text{cross-modal}}. \end{aligned} \tag{2}$$

Algorithm 1: MASKEDSCORE: log-likelihood of target tokens under a mask pattern

Input : sequence x; target positions P; extra masked positions X; vocabulary slice $[lo, hi)$ of the targets

Output : $\lambda_j = \log p_\theta(x_j \mid \cdot)$ and $\pi_j^{\max} = \max_y p_\theta(y \mid \cdot)$ for $j \in P$

Function *MASKEDSCORE*$(x, P, X, [lo, hi))$**:**

- $\tilde{x} \leftarrow x$; set $\tilde{x}_P \leftarrow \mathsf{M}$ and $\tilde{x}_X \leftarrow \mathsf{M}$;
- $\ell \leftarrow$ logits of $\tilde{x}$; ▷ `one network pass; batched with others`
- **for** $j \in P$ **do**
  - $\eta_j \leftarrow \mathrm{logsoftmax}_{\texttt{fp32}}\big(\ell_{j,[lo,hi)}\big)$;
  - $\lambda_j \leftarrow \eta_j[x_j - lo]$ and $\pi_j^{\max} \leftarrow \exp\max \eta_j$;
- **return** $(\lambda, \pi^{\max})$;

The quantity $q_j = B_j - D_j$ asks whether token j is consistent with the rest of its own modality, relative to how likely the value is at that position anyway. It never sees the other modality, and therefore never sees the hypothesis that will be tested. The cross-modal term $C_j - D_j$ is the decision evidence. All four levels come from one primitive, Algorithm 1.

3.3. The gate statistic and the evidence-separation rule

Two regimes are fixed per task before any run. In the regime with *no trusted reference* (detection, verification, matching), neither modality may be used to judge the other, because the cross-modal relation is what is being tested. In the regime with a *trusted reference* (repair with a known clean caption), the trusted modality may be used in the gate, as a predicted state is used in an innovation gate of tracking. The paper studies the first regime.

Evidence-separation rule: in the first regime, the gate may use only statistics computed with the other modality masked, that is B_j and D_j. A gate that used A_j with a candidate caption visible would give each candidate a different evidence set, and a gate that removed low cross-modal-evidence caption words would remove exactly the words that make a wrong caption wrong. The gate statistic is therefore

$$u_j = -q_j = -(B_j - D_j), \tag{3}$$

where a larger u_j means a more suspicious token. The division by the position marginal D_j is the optimal normalisation in one idealised case

only: if a corrupted value is drawn from the model's own position marginal independently of context, q_j is the Neyman–Pearson statistic, whereas for a uniform alternative, that is for corruptions that replace tokens by uniform random codes, the plain likelihood B_j is. Neither case is assumed to hold for JPEG, noise or splice, whose changed codes depend on the content, and no proposition about detection power is claimed for them. The null normalisation is therefore not assumed to improve detection, and the measurement does not show that it does. On the two uniform-code corruptions plain-B has a slightly higher mean per-item AUROC than q (patch 0.964 against 0.961, swap 0.960 against 0.955), and on JPEG and noise the two differ by less than 0.01 (0.560 against 0.561, and 0.511 against 0.503; no paired test was run on these two corruptions); Section 5.7 reports the full table. Detection of content-dependent damage is an empirical question (Section 4). The score $A_j - B_j$ without the null term cancels the own-context surprise of a corrupted token, because the term appears in both A_j and B_j; it is kept only as a control.

On the text side the gating unit is a word, that is a word-initial piece and its continuation pieces, and the suspicion of a word is the maximum of the suspicions of its pieces (equivalently, the minimum of their q), following the idea that multi-piece words need a word-level score [37]. Gating a word masks all its pieces.

3.4. Calibration by conformal risk control

The threshold is set on a calibration set $\mathcal{D}_{\mathrm{cal}}$ of n_{cal} clean pairs, disjoint by image from every other fold. "Clean" means images as distributed in the data set and passed through the tokenizer; corruption labels are relative to that tokenisation. For item i with m_i gating units and suspicions u_{ij}, the false-alarm proportion at threshold τ is

$$L_i(\tau) = \frac{1}{m_i} \sum_{j=1}^{m_i} \mathbb{1}\big[u_{ij} > \tau\big] \in [0, 1], \tag{4}$$

which is non-increasing in τ. Define the empirical calibration risk $\widehat{F}_{\mathrm{cal}}(\tau) = n_{\mathrm{cal}}^{-1} \sum_i L_i(\tau)$ and the threshold

$$\hat{\tau} = \inf \Big\{ \tau : \ \frac{n_{\mathrm{cal}}}{n_{\mathrm{cal}} + 1} \, \widehat{F}_{\mathrm{cal}}(\tau) + \frac{1}{n_{\mathrm{cal}} + 1} \leq \alpha \Big\}, \tag{5}$$

with $\hat{\tau} = +\infty$ if the set is empty, so that nothing is flagged. A token is flagged when $u_{ij} > \hat{\tau}$ with a strict inequality.

Algorithm 2: ReliabilityScores: suspicion of every token of one modality

Input : observed pair $\tilde{x}$; modality a; number of groups K; seed
Output : suspicion vector u over the positions of a

Function *ReliabilityScores*$(\tilde{x}, a, K, \text{seed})$**:**
  draw a partition Π_K of the positions of a from the seed;
  for $k = 1, \ldots, K$ **do**
    $B_{\Gamma_k} \leftarrow \textsc{MaskedScore}(\tilde{x},\ \Gamma_k,\ \text{positions of } b,\ \text{slice of } a)$; ▷ `b withheld`
  $D \leftarrow \textsc{MaskedScore}(\tilde{x},\ \text{all of } a,\ \text{all of } b,\ \text{slice of } a)$; ▷ `layout only; cached`
  $u_j \leftarrow -(B_j - D_j)$ for every position j of a;
  if $a = T$ **then** aggregate u to words by the maximum over the pieces of a word;
  return u;

Proposition 1 (calibration guarantee; conformal risk control applied)**.** *Let the calibration items and a new clean test item be exchangeable as items, and let L_i be defined by Eq.* (4)*, with scores that have no ties. Then* $\mathbb{E}\big[L_{\text{test}}(\hat{\tau})\big] \leq \alpha$.

Proof. The loss is bounded by one and non-increasing in the threshold, so the statement is Theorem 1 of conformal risk control [43] with bound $B = 1$. □

Wording and scope of the guarantee.. The statement made in this paper, and no stronger one, is: the threshold is calibrated so that, for a clean test pair exchangeable with the calibration pairs, the expected proportion of its tokens that are flagged is at most α. Three details keep the statement true in practice. The bound is over items, so tokens inside an item need not be exchangeable or independent and positions may have different score distributions. Ties, which arise from `bf16` logits, are broken by an independent uniform jitter $\epsilon \sim U(0, \epsilon_0)$ added to calibration and test scores, with ϵ_0 below the smallest non-zero gap between distinct calibration scores, computed and not chosen. The statistic, K, α and the fold sizes are fixed before any test item is scored, because choosing them from calibration or test items would break exchangeability. The guarantee is marginal over the draw of the calibration set, so one realised calibration set may lie above α on a finite test set.

Algorithm 3: CalibrateThreshold

Input : calibration set $\mathcal{D}_{\text{cal}}$; modality a; level α; K; seed rule
Data : clean pairs only, disjoint from all other folds
Output : threshold $\hat{\tau}$ and jitter scale ϵ_0, stored with the hash of the item list, K, seed and statistic name

Function *CalibrateThreshold*$(\mathcal{D}_{\text{cal}}, a, \alpha, K)$**:**
  for *each item* i *in* $\mathcal{D}_{\text{cal}}$ **do**
    $u_i \leftarrow \text{ReliabilityScores}(x_i, a, K, \text{seed}_i)$;
  $\epsilon_0 \leftarrow$ a value below the smallest non-zero gap between distinct pooled scores;
  add independent jitter $\epsilon_{ij} \sim U(0, \epsilon_0)$ to every u_{ij};
  evaluate $\widehat{F}_{\text{cal}}(\tau)$ at every distinct pooled value and return the smallest τ satisfying Eq. (5);

What is not guaranteed.. There is no bound on detection power: calibration controls false alarms on clean items only. There is no bound on false alarms among the clean tokens of a *corrupted* item, whose neighbours see contaminated context and can be flagged more often; that rate is measured and reported (Section 4). There is no guarantee under shift between calibration and test, such as another data set, caption style, resolution or tokenizer. Iterating the gate by re-scoring after masking breaks monotonicity in τ, so the bound would not apply to it. Words such as "certified" or "provably reliable" are not used for this gate.

3.5. Gating

The gate applies the frozen threshold to the suspicions of an observed pair. Algorithm 4 computes the gated set G and the reliable set R with the same K, seed rule and statistic as in calibration. No iteration is used.

3.6. Withholding flagged tokens: GATE-M (marginalise)

Masking a flagged token tells the network that its value is unknown. For a network that equals the Bayes-optimal denoiser of the absorbing-state objective, masking the set G marginalises it: for a masked target $j \notin G$,

$$p_\theta(x_j \mid x_R,\, x_G = \mathsf{M}) = \sum_{x_G} p(x_j \mid x_R, x_G)\, p(x_G \mid x_R) = p(x_j \mid x_R). \qquad (6)$$

Algorithm 4: GATE (uses Algorithms 2 and 3)

Input : observed pair $\tilde{x}$; modality a; frozen $\hat{\tau}$ and ϵ_0; K
Output : gated set G, reliable set R, suspicion u

Function $\textsc{Gate}(\tilde{x}, a, \hat{\tau}, \epsilon_0, K)$**:**
  $u \leftarrow \textsc{ReliabilityScores}(\tilde{x}, a, K, \text{seed})$;
  add independent jitter $\epsilon_j \sim U(0, \epsilon_0)$ to every u_j;
  $G \leftarrow \{j : u_j > \hat{\tau}\}$; $R \leftarrow$ complement of G;
  return (G, R, u);

This is the standard property of such denoisers [3], not a contribution. A real network approximates it. Structured patterns, such as a contiguous block of masked image tokens together with a fully masked caption, are rare under random per-token masking, so the amortisation error may be larger there; a coherence test on the real model is part of the protocol (Section 4).

When the set G is selected from the observed values, Eq. (6) is not enough. Let Y be the quantity to infer, for example the caption hypothesis. Using $p(Y \mid \tilde{v}_R)$ equals the Bayes answer under the corruption model when four conditions hold: (i) R contains no corrupted token; (ii) corrupted values carry no information about the clean values, which holds for replacement by uniform random codes and fails for JPEG and noise; (iii) corruption locations are independent of the content, which fails where changed codes concentrate in regions in which the tokenizer is unstable; (iv) no clean token is in G, or the clean tokens in G are uninformative about Y. The gate selects on the values of the tokens it withholds, so the missingness is informative [46]. Each violation has a cost: residual corrupted evidence in (i), discarded signal in (ii) because a noisy copy would call for a denoising posterior and not for marginalisation, a wrong distribution $p(v_G \mid v_R)$ for selected positions in (iii), and lost clean evidence at rate α in (iv). The conditions therefore favour withholding for corruptions that satisfy (ii) and (iii) and are least favourable, possibly not at all, for JPEG and noise; this is stated in advance and is not an account of any result.

3.7. Image-direction evidence

The decision studied is paired verification: for one image, does the true caption c^+ explain the image better than a wrong caption c^-. The image-direction evidence uses one pass per caption with all image tokens masked

and the caption visible. Let $\Lambda_j^c[y]$ be the log-probability of code y at image position j in that pass, so that

$$\Delta_j = \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j], \qquad \bar{\Delta}_R = \frac{1}{|R|}\sum_{j\in R}\Delta_j, \qquad Y_R = \mathbb{1}\big[\bar{\Delta}_R > 0\big]. \quad (7)$$

The position marginal D_j cancels in the difference for a fixed image, so $\Delta_j = (C_j(c^+) - D_j) - (C_j(c^-) - D_j)$. For a single candidate the per-caption evidence is $E_{I\leftarrow T}(c) = |R|^{-1}\sum_{j\in R}\big(\Lambda_j^c[\tilde{v}_j] - D_j[\tilde{v}_j]\big)$, and $\bar{\Delta}_R = E_{I\leftarrow T}(c^+) - E_{I\leftarrow T}(c^-)$. The logits of this pass do not depend on the image, which enters only through the gathered entries. Hence all arms that differ in the set R (no gate, gate, random gates, a changed-code gate, a denoised copy) reuse the same two passes per item. Ties and an empty set count as wrong, $Y = 0$. Two continuous summaries are used in the secondary analyses,

$$d_R = \frac{\bar{\Delta}_R}{s_R}, \qquad z_R = \sqrt{|R|}\, d_R, \quad (8)$$

where s_R is the sample standard deviation of Δ_j over R. The ratio d_R rises when tokens with extreme per-token evidence are removed for any reason, which is why a random gate and an uncertainty-ranked gate of equal size are reported next to it.

Analysis: when does withholding help an additive statistic?. Consider the statistic $S = \sum_j I_j\Delta_j$ over m tokens, with $I_j \in \{0,1\}$ the indicator that token j is kept. Under an independent-token approximation, clean tokens have mean μ and variance σ^2 and are kept with probability $1-\phi_0$, corrupted tokens have mean μ_c and variance σ_c^2 and are kept with probability $1-\phi_1$, and a fraction ρ of tokens is corrupted. Then

$$\mathbb{E}S = m\big[(1-\rho)(1-\phi_0)\mu + \rho(1-\phi_1)\mu_c\big], \quad (9)$$

$$\mathrm{Var}S = m\Big[(1-\rho)\big\{(1-\phi_0)\sigma^2 + \phi_0(1-\phi_0)\mu^2\big\} + \rho\big\{(1-\phi_1)\sigma_c^2 + \phi_1(1-\phi_1)\mu_c^2\big\}\Big]. \quad (10)$$

The no-gate case is $\phi_0 = \phi_1 = 0$, and the gate raises $\mathbb{E}S/\sqrt{\mathrm{Var}S}$ if the ratio after exceeds the ratio before. For $\mu_c = 0$ this reduces to

$$\rho\,\sigma_c^2\,(\phi_1 - 2\phi_0 + \phi_0^2) \;>\; \phi_0(1-\phi_0)(1-\rho)\,(\sigma^2+\mu^2). \quad (11)$$

Algorithm 5: ImageEvidence (GATE-M; uses Algorithms 1 and 4)

Input : observed image tokens $\tilde{v}$; candidate captions $c_1, \ldots, c_N$; reliable set R from Gate or another arm; cached position marginal D

Output : $E_{I \leftarrow T}(c_k)$ for every candidate and, for a pair (c^+, c^-), the paired quantities Y_R, $\bar{\Delta}_R$, d_R

Function *ImageEvidence*$(\tilde{v}, \{c_k\}, R)$**:**

  for *each candidate* c_k **do**

    $\Lambda^{c_k} \leftarrow$ MaskedScore with all image tokens masked and c_k visible, keeping the `fp32` log-probabilities over all image codes at all m positions; ▷ `one pass; reusable by every arm`

    $E_{I \leftarrow T}(c_k) \leftarrow \frac{1}{|R|} \sum_{j \in R} \left(\Lambda_j^{c_k}[\tilde{v}_j] - D_j[\tilde{v}_j] \right)$;

  if $N = 2$ **then** $\Delta_j \leftarrow \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j]$ for $j \in R$; **return** $\left(\mathbb{1}[\bar{\Delta}_R > 0],\ \bar{\Delta}_R,\ \bar{\Delta}_R / s_R \right)$;

  return $\{E_{I \leftarrow T}(c_k)\}$;

Three consequences are stated in advance. A necessary condition is $\phi_1 > 2\phi_0 - \phi_0^2$. A random gate has $\phi_1 = \phi_0$, for which the left side is negative, so it lowers the ratio. On a clean item ($\rho = 0$) any gate with $\phi_0 > 0$ lowers the ratio slightly. The model is an approximation, since neighbouring tokens are correlated, and μ_c need not be zero even for random codes: a caption that makes the predicted image distribution flatter gives random codes a higher probability. The protocol estimates $\rho, \mu, \sigma^2, \mu_c, \sigma_c^2$ and the observed flag rates $\hat{\phi}_0, \hat{\phi}_1$ per corruption from the per-token terms, and reports in how many corruptions the sign of the plug-in change of the ratio agrees with the sign of the observed net accuracy difference (descriptive family E-pred, Section 4.9); it is not a test and enters no claim about the primary result. E-pred uses Eqs. (9)–(10) as committed, and Proposition 2 below is neither used by it nor used to explain or interpret any E-pred result, agreeing or not.

Correlated tokens.. Eqs. (9)–(11) treat the per-token terms as independent, which is an approximation for image tokens. The following statement replaces independence by an exchangeable correlation within and between the two classes of tokens. It is theory under stated assumptions and is not evaluated on the model. The new symbols are γ_0, γ_1 and γ_{01}, the correlations of two clean, two corrupted, and one clean and one corrupted token.

Proposition 2 (withholding with exchangeably correlated tokens)**.** *Condition on the corruption labels, so that* $(1-\rho)m$ *tokens are clean and* ρm *are corrupted, with the moments* $\mu, \sigma^2, \mu_c, \sigma_c^2$ *and the keep probabilities* $1-\phi_0$*,* $1-\phi_1$ *of Eqs.* (9)–(10)*. Assume (a) that for* $j \neq k$ *the correlation of* Δ_j *and* Δ_k *is* γ_0 *if both tokens are clean,* γ_1 *if both are corrupted and* γ_{01} *otherwise, with values for which the covariance matrix is positive semi-definite; and (b) that the keep indicators* I_j *are independent of each other and of every* Δ_k*. Then* $\mathbb{E}S$ *is given by Eq.* (9)*, and, under (a) and (b),*

$$\begin{aligned} \mathrm{Var}S = V &+ (1-\rho)m\big((1-\rho)m-1\big)(1-\phi_0)^2\gamma_0\sigma^2 \\ &+ \rho m(\rho m-1)(1-\phi_1)^2\gamma_1\sigma_c^2 \\ &+ 2\rho(1-\rho)m^2(1-\phi_0)(1-\phi_1)\gamma_{01}\sigma\sigma_c, \end{aligned} \tag{12}$$

where V *is the right side of Eq.* (10)*. If moreover* $\mu_c = 0$ *and* $\mathbb{E}S > 0$ *with and without the gate, the gate raises* $\mathbb{E}S/\sqrt{\mathrm{Var}S}$ *if and only if*

$$\begin{aligned} \rho\,\sigma_c^2\Big[&(\phi_1 - 2\phi_0 + \phi_0^2) \\ &+ (\rho m - 1)\,\gamma_1\,(\phi_1-\phi_0)(2-\phi_0-\phi_1)\Big] \\ + 2\rho(1-\rho)\,m\,&\gamma_{01}\,\sigma\sigma_c\,(1-\phi_0)(\phi_1-\phi_0) \\ > \;&(1-\rho)\,\phi_0(1-\phi_0)\,(\sigma^2+\mu^2). \end{aligned} \tag{13}$$

Consequently: (i) γ_0 *does not enter Eq.* (13)*, and at* $\gamma_1 = \gamma_{01} = 0$ *it is Eq.* (11)*; (ii) a gate with* $\phi_1 = \phi_0$*, such as a random gate, lowers the ratio for every admissible correlation, and if* $\gamma_{01} \geq 0$ *every gate with* $\phi_1 \leq \phi_0$ *lowers it; (iii) where* $\phi_1 > \phi_0$*, a larger* γ_1 *or* γ_{01} *can only turn a decrease of the ratio into an increase, so positive correlation among corrupted tokens, or between clean and corrupted tokens, enlarges the region of* (ϕ_0, ϕ_1) *in which the gate raises the ratio, and as* $(\rho m - 1)\gamma_1 \to \infty$ *that region tends to* $\{\phi_1 > \phi_0\}$*; (iv) if* $\gamma_{01} < 0$*, a gate that flags only clean tokens can raise the ratio; (v) unlike Eq.* (11)*, the condition depends on* m*.*

Proof sketch. Each of the $m(m-1)$ ordered pairs of distinct tokens adds $\mathbb{E}I_j\,\mathbb{E}I_k\,\mathrm{Cov}(\Delta_j, \Delta_k)$ to $\mathrm{Var}S$, which gives the three correlation terms of Eq. (12). With $\mu_c = 0$ the gate multiplies $\mathbb{E}S$ by $1-\phi_0$, so it raises the ratio if and only if $(1-\phi_0)^2\,\mathrm{Var}S|_{\phi_0=\phi_1=0} > \mathrm{Var}S$; the clean-pair terms scale by $(1-\phi_0)^2$ on both sides and cancel, and the remaining terms, divided by m, give Eq. (13). Appendix Appendix D gives the full proof. □

At zero correlation the statement is the independent-token analysis above: Eq. (12) reduces to Eq. (10) and Eq. (13) to Eq. (11). Correlation therefore moves the gating frontier in the way (ii)–(iv) state, and independence is an approximation whose effect on the frontier is of that kind and not of a fixed sign. The ratio in Proposition 2 is the signal-to-noise ratio of the sum. If the per-token terms are jointly Gaussian, the accuracy $P(S > 0)$ given the flags is Φ of the conditional ratio, which the unconditional ratio approximates; without joint Gaussianity no statement about accuracy follows, and a gate can raise the ratio while lowering the accuracy. Assumption (b) is an approximation for GATE, whose flags are computed from the same token values as the evidence. With correlation the ratio no longer grows like $\sqrt{m}$: it tends to a limit set by the shared component of the per-token terms.

3.8. Text-direction evidence and conditioning arms

The text direction scores the caption tokens given the image. With caption c fully masked and the image visible, the evidence is the sum over caption tokens of the log-probability minus the same pass with the image masked,

$$E_{T \leftarrow I}(c) = \sum_{j} \big(C_j - D_j\big), \tag{14}$$

where the image tokens in the gated set G are masked in the first pass. This is a conditioning arm in which the network, and not the arithmetic of a sum, withholds the flagged tokens. In the exploratory analyses the statistic is the mean per-token value.

3.9. Re-infilling flagged tokens: GATE-I (impute)

Instead of leaving flagged tokens masked, the model can fill them in. Algorithm 6 samples n_{draw} completions in S_r steps, unmasking at step s the next k_s positions of a random order drawn up front, and keeps the draw with the highest chain log-likelihood under the model itself, with the caption masked (best of n_{draw}; GATE-I). The sampler is unguided and samples each position from the softmax at temperature one; it is the sampler fixed in the analysis plan and not the sampler of the backbone's authors. A hybrid variant that keeps only filled values whose recorded probability reaches κ is described in the specification and is not part of any preregistered family. If the filled tokens are generated with the candidate caption visible, the evidence for that

Algorithm 6: ReInfill (GATE-I; uses Algorithms 1 and 4)

Input : observed image tokens $\tilde{v}$; gated set G; steps S_r; draws n_{draw}; seed

Output : the kept completed image $\hat{v}$ and the n_{draw} draws

Function $\textit{ReInfill}(\tilde{v}, G, S_r, n_{\text{draw}}, \text{seed})$**:**

  for $\ell = 1, \ldots, n_{\text{draw}}$ **do**

    $y \leftarrow \tilde{v}$ with $y_G \leftarrow \mathsf{M}$; caption masked ; ▷ `caption masked: no leakage`

    draw a random order of G from the seed; $\text{chain}_\ell \leftarrow 0$;

    for $s = 1, \ldots, S_r$ **do**

      $U \leftarrow$ positions of G still masked in y;

      $k_s \leftarrow \lceil |U|/(S_r - s + 1) \rceil$;

      $\pi_j \leftarrow p_\theta(\cdot \mid y)$ for $j \in U$; ▷ `one pass, via MaskedScore`

      take the next k_s positions of U in the drawn order; sample $y_j \sim \pi_j$; $\text{chain}_\ell \mathrel{+}= \log \pi_j(y_j)$;

    $\hat{v}^{(\ell)} \leftarrow y$;

  return the draw $\hat{v}^{(\ell)}$ with the largest chain_ℓ and the set of all draws;

caption is biased upward. Writing $\hat{v}_G \sim p(v_G \mid v_R, c)$, Bayes' rule gives

$$\begin{aligned} \mathbb{E}\big[\log p(c \mid v_R, \hat{v}_G)\big] &= \log p(c \mid v_R) + \mathrm{KL}\big(p(v_G \mid v_R, c) \,\|\, p(v_G \mid v_R)\big) \\ &\geq \log p(c \mid v_R), \end{aligned} \tag{15}$$

so every candidate receives a non-negative bonus equal to how strongly it constrains the filled region, which need not favour the true caption. The rule is therefore to re-infill with the caption masked and to run the caption-visible variant only as a leakage control. With exact sampling from $p(v_G \mid v_R)$, the expectation of $p(c \mid v_R, \hat{v}_G)$ equals $p(c \mid v_R)$, so GATE-I is unbiased for GATE-M in the probability domain; in the log domain an n_{draw}-sample estimate is biased downward by Jensen's inequality and adds sampling variance. Marginalising and imputing are not compared as a test: the preregistered re-infill family S2 compares the repaired flagged set with repairing an equal-size random set (Section 4.9).

3.10. Two-way evidence fusion

For a candidate c there are two directional statistics, $E_{T\leftarrow I}(c)$ of Eq. (14) and the image-direction evidence of Eq. (7). Inverse-variance weighting is not used, because the two statistics estimate different quantities on different scales, share one network and are deterministic for a given input under full masking. The fusion rule is reported in three steps on the paired differences $\Delta^{\text{pair}} = (\Delta_T, \Delta_I)^\top$ with $\Delta_d = E_d(c^+) - E_d(c^-)$: (1) each direction alone; (2) an unfitted Stouffer sum, with each paired margin standardised by its standard deviation on calibration items and the sum divided by $\sqrt{2 + 2\hat{r}}$, where $\hat{r}$ is the calibration correlation; (3) Fisher's linear discriminant,

$$\hat{\omega} = \hat{\Sigma}_\Delta^{-1}\hat{\mu}_\Delta, \qquad F(c) = \hat{\omega}_T\, E_{T\leftarrow I}(c) + \hat{\omega}_I\, E_{I\leftarrow T}(c), \tag{16}$$

with the weights fitted on pairs disjoint from the test pairs and frozen before the test fold. The fusion therefore has two fitted weights, which is why the paper says "no network finetuning" and not "training-free". The sum of per-token log-likelihood ratios in Eq. (7) is the fusion rule of distributed detection [8], which equals the global log-likelihood ratio only for sources that are independent given the hypothesis. Image tokens are not independent, so independence is an approximation here; Proposition 2 states how correlation moves the gating frontier, and it does not make the sum optimal. The sum is used as a standard statistic and not as an optimal one. Whether a fusion gain can be tested depends on the clean accuracy of the best single direction (Section 4).

4. Experimental protocol

The protocol was fixed before any outcome of the main run existed. It is recorded in a preregistration card (version 4), a statistical analysis plan (SAP) with its addenda, and a method specification, all pinned by commit and hash (Section 4.15 and Appendix A). Eleven test families are preregistered, each at its own level with no control across families. Only the primary family P can carry the headline: no secondary family can rescue or explain a null result in P, and none is stated in the title, abstract, highlights or conclusion without its family label. This section states what is run, with what, and how each result will be judged; parts of the method that a family does not exercise are named as such.

Algorithm 7: GATEVERIFY: gated cross-modal verification (composes Algorithms 1–6)

Input : observed pair $\tilde{v}$ and candidates $c_1, \ldots, c_N$; frozen $\hat{\tau}_I, \epsilon_0$; fusion weights $\hat{\omega}$; arm $\in$ {none, GATE-M, GATE-I}

Output : a score $F(c_k)$ per candidate and a decision

Function *GATEVERIFY*$(\tilde{v}, \{c_k\}, \hat{\tau}_I, \hat{\omega}, arm)$**:**

  $(G, R, u) \leftarrow$ GATE$(\tilde{v}, I, \hat{\tau}_I, \epsilon_0, K)$; ▷ `once per image; caption masked, hypothesis-free`

  if *arm* = *none* **then** $G \leftarrow \varnothing$; $R \leftarrow \{1, \ldots, m\}$;

  if *arm* = *GATE-I* **then**

    $\hat{v} \leftarrow$ REINFILL$(\tilde{v}, G, S_r, n_{\text{draw}}, \text{seed})$; $\tilde{v} \leftarrow \hat{v}$; $R \leftarrow \{1, \ldots, m\}$

  $\{E_{I \leftarrow T}(c_k)\} \leftarrow$ IMAGEEVIDENCE$(\tilde{v}, \{c_k\}, R)$;

  for *each candidate* c_k **do**

    $E_{T \leftarrow I}(c_k) \leftarrow$ MASKEDSCORE on the caption, caption fully masked, image visible with the tokens of G masked (GATE-M);

    $F(c_k) \leftarrow \hat{\omega}_T E_{T \leftarrow I}(c_k) + \hat{\omega}_I E_{I \leftarrow T}(c_k)$;

  return $\{F(c_k)\}$ and the decision (argmax, or a threshold for unpaired verification);

4.1. Model, data and tokenisation

The model is UniDisc with 1.4B parameters [4], used as released, with no training or finetuning here. Images and captions come from the COCO val2017 split [48]. Images are tokenised at 512 px with the VQ-16 tokenizer of LlamaGen [49], giving a 32×32 grid of $m = 1024$ image tokens drawn from 16 384 codes. Captions of 5 to 32 text tokens are used. Instance annotations of val2017 are used to restrict wrong captions (Section 4.2). Whether COCO val2017 images were in the training data of UniDisc or of the tokenizer is unknown to the authors and is a stated limitation (Section 6). The backbone runs in `bf16`; log-probabilities are an `fp32` log-softmax of the logits restricted to the scored modality.

Time input.. UniDisc's denoiser has no time input in the released configuration: time conditioning is off, the official inference configuration passes a null noise level, the backbone builds no timestep embedder without time conditioning, and the released checkpoint contains no time-embedding weights;

the scorer uses the authors' null-sigma inference path. This is a property of the released configuration and checkpoint, read in the authors' code at the pinned commit (files and lines in Appendix B), and not a statement about UniDisc in general. Because the loader is non-strict, a key check was run: all 342 checkpoint tensors match the 342 parameters of the denoiser backbone, with no missing key, no unexpected key and no shape mismatch. The check does not cover the image tokeniser and does not show that the values are the intended final weights, and how often whole-modality masks occurred in training cannot be recovered from the code.

4.2. Folds, negatives and tokeniser check

Four disjoint sets of images are drawn by COCO image id with a fixed seed (20261029): calibration (300 items), headroom (100), test (300, with an ordered extension to 500 used only if the sample-size rule of Section 4.4 asks for it), and the 200 items of an earlier detection pilot, which serve as the detection set. All new folds exclude the 494 image ids of the two pilots. The wrong caption c^{-} of an item is a caption of another image, fixed once per item and shared by every arm and corruption; where annotations exist, it comes from an image that shares no annotated COCO category with the true image. Before any score is computed, the frozen caption token counts are compared with the counts of the runtime tokeniser. An item whose real count falls outside 5 to 32 is excluded from every analysis, counted and not replaced; if more than 10 % of any fold fails the check the run stops and a fold-construction defect is reported.

4.3. Corruptions

Table 3 lists the corruptions. Three are *realistic*: they act on pixels before tokenisation and produce plausible codes. Two are *sanity* corruptions that replace tokens by uniform random codes; they form the separate positive-control family PS, not the primary family. The corrupted-token label is $Z_j = \mathbb{1}[\tilde{v}_j \neq v_j^{\circ}]$, relative to the tokenisation of the clean image. Noise draws and the donor image and location of a splice derive from the item id, so one corruption realisation exists per image.

4.4. Headroom rule and sample size

A gate can restore verification only if corruption lowers it and removing the corrupted tokens recovers it. Both are checked on the 100 headroom items, blind to the method: no gate flag is computed on those items. The endpoint

Algorithm 8: GEOMETRYMATCHEDGATE: a random gate with the shape of the gate

Input : gated set G on the 32×32 grid; seed
Output : a random gate G^{geo} of the same size and shape at another place

Function *GEOMETRYMATCHEDGATE*(G, seed)**:**
  draw (d_x, d_y) uniformly from $\{0, \ldots, 31\}^2 \setminus \{(0,0)\}$;
  return $\{((x + d_x) \bmod 32,\ (y + d_y) \bmod 32) : (x, y) \in G\}$;
  ▷ `torus shift`

is the size-normalised evidence d_R of Eq. (8). For corruption c and rung r let $u_i = d_i^{\text{clean}} - d_i^{\text{cor}}$ and $v_i = d_i^{\text{chg}} - d_i^{\text{cor}}$, where "chg" is the changed-code gate that keeps exactly the tokens that did not change. With T_1 and T_2 the paired t statistics of u and v over the n_v items that have all values, a rung passes if $n_v \geq 90$, $T_1 \geq 2.0$ and $T_2 \geq 2.0$. The first passing rung of a corruption is its operating point; a corruption that passes neither rung has none, is not tested, and keeps a p-value of one in the primary family. If no corruption has an operating point, the downstream claim is withdrawn and the paper reports detection, calibration and the clean control only.

The headroom items also fix the test-fold size. Let D_c be the number of headroom items whose paired correctness differs between the changed-code gate and no gate, and $D^* = \max_c D_c$ over corruptions with an operating point. Rule R1 ($D^* \geq 29$): test on the first 300 ids with the correctness endpoint. Rule R2 ($18 \leq D^* < 29$): test on the first 500 ids. Rule R3 ($D^* < 18$): test on the first 300 ids with the paired change of d_R as the primary endpoint and correctness as a secondary one. The decision, D_c and D^* are written to a file that is committed before the first test item is scored.

4.5. Arms and controls

Table 4 lists the arms of the image-direction analysis. All arms reuse the two caption-only passes per item (Table C.20). The primary contrast is GATE-M against *none*. The controls separate three competing explanations: mask geometry (an equally shaped mask elsewhere), texture or encoder instability (cheap non-generative detectors), and pixel denoising.

4.6. Detection baselines and comparators

Detection of changed codes is evaluated against four non-generative or uncalibrated rivals, fixed in advance: the *patch texture* (pixel variance of a token's 16×16 patch), the *neighbour codebook distance* (mean L_2 distance of the token's codebook vector to its eight neighbours), the *code rarity* (negative log frequency of the code among the calibration fold's clean codes) and the *uncertainty score* s_3. Both signs are admitted for each rival, which favours the rival (a sign oracle), and the best rival is chosen on the data. Plain-B is the detector of Eq. (3) without the null normalisation, and $A_j - B_j$ is the cancellation control (expected near 0.5). The *TbV-score (non-finetuned)* is the observed-position probability of Trust but Verify [28] computed from the raw backbone logits of the unmodified model with the caption masked, never from the released output layer, which writes a probability of one at every unmasked position; it reproduces that paper's score but not its method, which needs a finetune, and it is reported threshold-free. The random-gate best-of-four re-infill (I-RAND4) of family S2 is the closest we can build to the repair demonstration of the backbone's authors [4]: an equal-size random set is re-infilled four times and the draw with the highest chain likelihood under the model is kept, with the sampler of the analysis plan, so it says nothing about the backbone's own re-infilling. The *pixel denoiser* is OpenCV non-local-means with one setting for all corruptions (strength matched to noise level 0.10). On 20 images it raises the PSNR of the noisy image at $\sigma = 0.10$ from 20.41 to 25.75 dB and lowers that of JPEG quality 10 from 26.82 to 25.13 dB, so on JPEG GATE-M is compared with an aggressive denoiser that harms the image, and the JPEG row is worded accordingly. Token-Critic [25], CO2Jump [27] and FreeCorrection [26] are discussed but not run: they need a trained critic, are sampler-level methods for generation, or have code that is not runnable at the pinned commit.

4.7. The primary family and how it is read

Primary family.. For each of JPEG, noise and splice, at the operating point from the headroom rule, the endpoint is the paired correctness Y of Eq. (7) under GATE-M against none. The test is the exact two-sided McNemar test [50] on the discordant items: with b items right under GATE-M and wrong under none, c items the reverse, and $n_d = b + c$,

$$p = \min\Big\{1,\ 2\,\mathbb{P}\big(\mathrm{Bin}(n_d, \tfrac{1}{2}) \le \min(b, c)\big)\Big\}. \tag{17}$$

The family has exactly three hypotheses at all times, with Holm step-down [51] at familywise level 0.05: a corruption without an operating point stays in the family with $p = 1$. With every test the paper reports b, c, n_d, n, both accuracies, the net difference $(b - c)/n$ with paired bootstrap intervals (10 000 resamples) at 95 % and at 98.33 %, and the Clopper–Pearson interval [52] of b/n_d. The design minimum detectable net gain at $n = 300$ is 0.053 unadjusted and 0.060 at the first Holm step, under an assumed discordance of 10 % (assumed, not observed).

Reading of the primary result.. The wording for every outcome was fixed before the headroom decision of the main run was written, in an outcome book of the analysis plan (Addendum A4), and the title, highlights and abstract follow it. The outcome is labelled *positive* if at least one hypothesis is rejected after Holm with $b > c$ and none with $c > b$; it supports only "a gate built on the masked-likelihood score improved image-direction verification over no gate on that corruption". Attribution to detection needs the mechanism family: if GATE-M against the geometry-matched random gate is not rejected, the improvement is "not distinguishable from masking the same shape elsewhere, and therefore not attributable to detection beyond mask geometry". A rejection with $c > b$ is reported as "GATE-M reduced verification accuracy", and no claim in favour of the gate is made from it; with rejections of both kinds the outcome is *mixed*. The outcome is *negative* if no hypothesis is rejected and, for every corruption with an operating point, the 90 % paired bootstrap interval of the net accuracy difference lies inside $(-0.05, +0.05)$ with at least 240 completed items; it is *inconclusive* otherwise. A null result is worded "no improvement was detected; the data are compatible with a net accuracy change between L and U", and never "GATE-M has no effect". Under rule R3 the primary endpoint is the evidence ratio, the label negative is not available, and the positive sentence says that GATE-M increased the evidence ratio and nothing about verification accuracy, with the caveat that removing heavy-tailed tokens can raise a ratio for any reason. If no corruption has an operating point, the outcome is "no operating point" and no statement about gating and verification is made.

Rules that stop or reword a claim.. These are fixed in advance. The improvement claim is withdrawn if no hypothesis is rejected, and the null wording above applies. The claim of a working guarantee is suspended if the calibration statistic Z_{cal} exceeds 2.13 at any level, until an audit against the unit

tests ends. An arm with more than 5 % empty or non-finite evidence stops and is diagnosed from logs. If the score does not beat the best cheap rival on both JPEG and noise, the paper may not say that it detects corrupted codes better than cheap methods, and it reports the rival, its sign and its AUROC with an interval in the main text; the primary test still runs. A failed positive-control chance bar does not block the primary test; it triggers an audit and is reported as a pipeline defect.

4.8. Why the image direction is the primary endpoint

The primary decision uses the image-direction statistic, which is the weaker of the two statistics available on clean pairs. In the exploratory pilot (Section 4.13) the full-mask text direction reached 0.990 and the image direction 0.843. The image direction became the primary endpoint because it has headroom and because the corruptions studied act on the image. The choice was made after seeing the pilot. A practitioner who wanted to verify captions would use the text direction, or fuse the two. The primary result therefore describes how well the weaker statistic tolerates image damage, with and without a gate, in a setting where the wrong caption is a caption of another image that shares no annotated category with the true one, an easy matching task. It does not show what a gate buys a verifier that already uses the stronger statistic. The only preregistered test of the two-direction setting is family B (Section 4.12), on harder negatives, and it is declared untestable if the better single direction is already at or above 0.95 accuracy on the fit pairs. If B is untestable or is not run, the question of what gating adds to a near-ceiling verifier is left open.

4.9. Secondary and descriptive families

Besides P there are ten test families; Table A.17 in Appendix A gives their sizes, tests and wording rules. *PS* is the positive control, the same test on the two sanity corruptions with Holm over two: a pass means "the pipeline is not broken", a failure triggers an audit, and it is never shown beside P as a demonstration that the gate works where detection is easy. *S2* tests re-infilling on patch and splice, the repaired flagged set against the repaired equal-size random set (I-GATE against I-RAND4), and only if at least 120 items are complete. *B* tests two-way fusion on hard negatives (Section 4.12). *R* repeats the primary test on a second model, a unified masked diffusion model, Omni-Diffusion [53], of 8B parameters quantised to 4 bits **[AUTHOR: the cited paper does not state the size or the quantisation; match**

this wording to the checkpoint in .venv-omni], on 150 test images, and runs only if P has a rejection with $b > c$. Six further families come from the first version of the plan: mechanism (S-M), evidence ratio (S-Z), denoising (S-DN), detection (S-D, with its out-of-sample version S-D2) and the clean control (S-C). E-dec, E-pred and an exploratory corruption suite (the imagecorruptions package [54]) are descriptive and have no test. E-dec decomposes the accuracy lost to corruption into the residual loss under the changed-code (oracle) gate and the gap to that oracle; E-pred reports in how many of K_{eval} cells the sign of the plug-in signal-to-noise change of Eq. (11) agreed with the sign of the observed net accuracy difference. Each family has its own level 0.05 and Holm correction, with no multiplicity control across families. Families run in a fixed order, and anything unfinished at the results freeze is cut in the reverse order by time only (the suite, then R, B and S2 last), declared not run and never replaced.

4.10. Calibration diagnostics and pipeline checks

On the clean test items the realised false-alarm proportion $\hat{F} = n^{-1} \sum_i L_i(\hat{\tau})$ is reported for $\alpha \in \{0.01, 0.05, 0.10\}$, with $\alpha = 0.05$ primary; a calibration statistic Z_{cal} above 2.13 at any level is a violation flag, and a pass means "no evidence of violation", never validation. The precision among flagged tokens and the false-alarm proportion on the clean tokens of corrupted items are reported with equal prominence. A positive-control chance bar on the sanity corruptions and a coherence test of the denoiser across mask patterns serve as pipeline checks; their definitions and readings are in Appendix C.

4.11. Exploratory comparison with robust and weighted combiners

The gate gives each token a weight of zero or one, whereas the usual answer to unreliable sources is graded: weights that follow reliability [8, 14], or aggregators that need no detector, such as a median or a trimmed mean. The hard weight was chosen because the conformal threshold bounds the expected share of flags on clean items, which a graded weight does not; it was not shown to be better. With dependent image tokens, whether the detector adds anything beyond robust aggregation is a fair question. SAP Addendum A4, committed before the headroom decision of the main run was written, pre-specifies an exploratory comparison on the stored per-token evidence Δ_j of the primary test fold, with the primary decision rule. GATE-M is compared with five combiners of the same Δ_j: the median; a trimmed mean that removes

as many tokens as the gate flagged for the item, half from each tail; a 20 % trimmed mean; a discount of the flagged tokens whose weight falls with their suspicion; and a weight on every token equal to the conformal p-value of its suspicion against the clean calibration tokens. Paired accuracy differences are reported with bootstrap 95 % intervals and the minimum detectable difference at the realised spread; no p-value is computed and no multiplicity adjustment is made. A detector-free combiner (the median or either trimmed mean) whose accuracy is above that of the gate counts against the method; one that is not distinguishable counts against the added value of the detector only if the minimum detectable difference is at most 0.05, and is otherwise uninformative at the realised n. For the two graded weights, a higher accuracy counts against discarding tokens, not against the detector. The comparison is exploratory (Tier 3): it has its own Results subsection after the preregistered families, it is absent from the title, abstract, highlights, conclusion and contributions, and it cannot explain, qualify or rescue a primary result. It was run once, after the primary analysis was committed (Section 5.14), with the direction of every contrast read from an exact interval for the discordant items (Addendum A4, revision r3).

4.12. Two-way fusion and the text direction (family B)

Family B asks whether the score-level fusion of the two directions, with weights fitted on separate fit pairs (Eq. (16)) and frozen, is more accurate than the better single direction on a hard-negative set built from FOIL-COCO [55] (exact McNemar, one hypothesis). It is analysed only if the better single direction has an accuracy below 0.95 on the fit pairs; otherwise the paper states that the fusion gain cannot be tested at that ceiling and gives the clean accuracy. B is not a corruption test and says nothing about corruption. The text-direction conditioning arms (none, GATE-M with the flagged image tokens masked, geometry-matched, changed-code) run for the three realistic corruptions and are exploratory. Caption corruption is not part of any family, and the paper makes no statement about it.

4.13. Exploratory feasibility result

A pilot on 300 clean COCO images, run before the protocol was fixed, shows the pair-level signal on which the endpoint rests. For each image the true caption was compared with a caption of a different image; accuracy is the share of images for which the true caption scores higher (chance 0.5; bootstrap intervals over images, 10 000 resamples). With $K = 8$ complementary masks

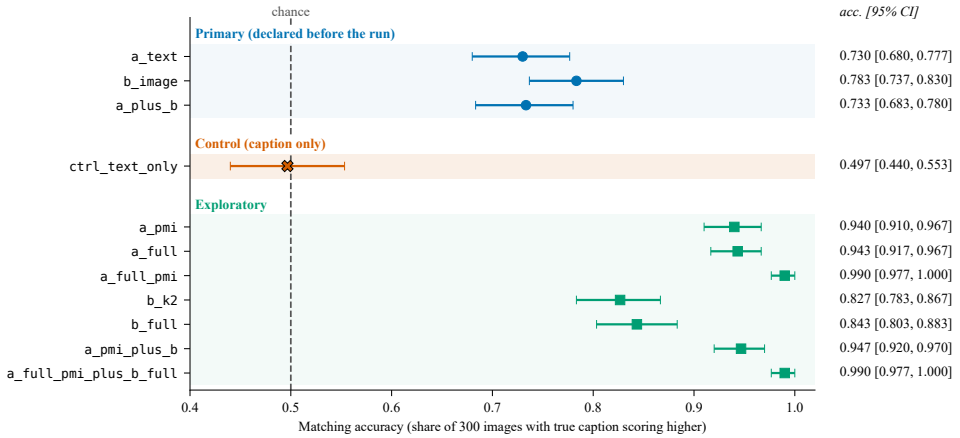


Figure 1: Matching accuracy of UniDisc 1.4B on true versus mismatched COCO captions (exploratory feasibility result, 300 images). Markers show the share of images for which the true caption scores higher (chance 0.5, dashed); bars are 95 % bootstrap intervals over images (10 000 resamples). Blue: primary scores, text direction a_text and image direction b_image with $K = 8$ complementary masks, and their sum. Orange: caption-only control, which stays at chance. Green: variants chosen after the primary results (image-masked null subtracted, single full-mask pass, $K = 2$, sums). One run, one mismatch draw; the intervals do not cover variation across draws or models.

the text direction reaches 0.730 (95 % interval 0.680 to 0.777) and the image direction 0.783 (0.737 to 0.830), and a caption-only control with the whole image masked stays at chance, 0.497 (0.440 to 0.553). Variants chosen after seeing these results, with all tokens of the scored modality masked in one pass and the image-masked null subtracted, reach 0.990 (0.977 to 1.000) in the text direction and 0.843 (0.803 to 0.883) in the image direction (Figure 1), whose mean paired margin is 0.18 nats per token. This is one run with one mismatch draw, on clean images, with the full-mask variants selected post hoc; it is exploratory and carries no inference beyond the intervals shown.

4.14. Hardware, runtime and missing items

All runs use one laptop GPU (12 GB) in short checkpointed chunks under a guard (Appendix C). Missing items are handled by a fixed rule: complete cases per analysis, no replacements and no reordering, so a crash leaves a prefix of the item list; if the completed base fold has fewer than 240 items the paper says so, prints the minimum detectable effect at the achieved n, and does not read "not rejected" as evidence against GATE-M. No analysis that computes Y, b, c, p or a difference of d_R runs on test data before the

planned fold is complete. Mask-seed and corruption-seed variance are not in the intervals.

4.15. Preregistration, pins and disclosed deviations

The code, the tests, the fold file, the hard-negative pool and the analysis plan are pinned by commit and SHA-256, and the pins, the audit and sign-off record, the deviations known at the time of writing and the dated amendments are in Appendix A. Changes after a pin are logged as dated amendments; none is made inside a stage. One analysis preceded the plan's second addendum and bears on the detection wording. The first 21 items of the detection set were analysed on CPU without the gate statistic q, which needs a GPU pass. In that exploratory look the mean per-item AUROC against changed-code labels of plain-B was 0.54 on JPEG quality 10 and 0.51 on noise 0.10, while pixel variance with its sign flipped scored 0.76 and 0.77. The sign of the pixel-variance rival therefore cannot be claimed to be fixed blind, which is why both signs of every rival are admitted and the rival is chosen on the data. **[AUTHOR: final list of amendments and the date of the first M2 outcome; confirm the status of the open decisions on guard thresholds, the JPEG denoiser setting and the fallback before the run.]**

5. Results

Conventions for this section.. Every interval is stated with its level. Unless a caption says otherwise, intervals are paired percentile bootstrap intervals over items with 10 000 resamples, and accuracy means image-direction verification accuracy, the share of items whose true caption has the larger image-direction evidence (Eq. (7); higher is better). One emphasis rule holds for every table of this section: an estimate is set in bold when the interval in its row excludes zero (for the exploratory accuracy contrasts of Tables 13 and 14, when the exact interval for the discordant split excludes one half); nothing else is bold and nothing is underlined. Every caption names the file and key that hold its numbers.

5.1. Primary result: GATE-M against no gate (family P)

In the preregistered test no improvement was detected under JPEG quality 10, Gaussian noise with $\sigma = 0.10$ or a 128-pixel splice. For each tested corruption the 90 % interval of the net accuracy change lay inside

± 0.05: -0.002 (-0.006 to 0.000) under JPEG, 0.000 (0.000 to 0.000) under noise and -0.004 (-0.010 to 0.002) under splice, with $n = 500$ each, so an improvement larger than 0.05 is excluded and a loss larger than 0.05 is excluded. This is not a finding that GATE-M has no effect. It holds in one pretrained unified masked diffusion model (UniDisc, 1.4B parameters) on COCO image-caption pairs with a random other caption as the negative, at one severity per corruption. By the pre-stated rule (Section 4.7) the label of the primary family is *negative*: no hypothesis was rejected after Holm correction over three, and every corruption had an operating point and 500 completed items.

Table 5 gives the full test. The gate changed the decision on one of 500 items under JPEG, on none under noise and on four under splice, where it was right on one and wrong on three ($b/n_d = 0.25$, exact 95 % interval 0.006 to 0.806). Under noise every bootstrap interval is the single point zero because no item was discordant; it records the absence of discordant items, not a measured precision. All 500 planned items of the test fold were completed, no item failed the caption-token check, and no arm had empty or non-finite evidence.

[RESULT FIGURE (A4.2.5, family P only): one row per corruption, net accuracy difference GATE-M minus none with the 90 % (thick) and 95 % (thin) bootstrap intervals, a solid line at 0 and dashed lines at ± 0.05; nothing else on the canvas. Data: `runs/P2-GATE-M4/analysis.json`, `primary.per_corruption.*.{delta,ci90,ci95}`**.]**

Operating points and sample size.. The operating points and the fold size were fixed on the 100 headroom items before any test item was scored (Section 4.4). The first rung of every corruption passed both headroom conditions (Table 6): JPEG quality 10, noise $\sigma = 0.10$ and the 128-pixel splice. The largest discordance between the changed-code gate and no gate among these rungs was $D^* = 19$, which selects rule R2: 500 test items with the correctness endpoint. On the headroom items JPEG quality 10 changed 986.7 of the 1 024 codes of an image on average, noise 949.4 and the splice 237.2, and the splice lowered the clean accuracy of the 100 headroom items by only 0.01. On the size-scaled ratio z_R of Eq. (8), a sensitivity analysis, the second condition fails at every rung (T_2 from -6.75 to -0.09); the rule uses d_R.

5.2. Mechanism controls (family S-M) and equal-size gates

Family S-M tests GATE-M against the geometry-matched random gate only on a corruption with a primary rejection with $b > c$; the others keep $p = 1$ in its Holm family of three. There was no such corruption, so S-M tested no hypothesis, and its two pre-stated rules for an improvement that is not attributable to detection do not apply. The contrasts of GATE-M with the other gates of equal size are reported descriptively in Table 7, with paired bootstrap intervals and no p-value. On every realistic corruption GATE-M was not distinguishable from the geometry-matched and the scattered random gates, each averaged over its three seeds. Under splice the top-k gate on plain B, which removes the same number of tokens with the lowest uncalibrated likelihood, gave a higher accuracy than GATE-M (-0.008, 95 % interval -0.016 to -0.002). The changed-code gate, which keeps only the codes that the corruption left unchanged, lowered accuracy relative to no gate under JPEG (-0.084, -0.122 to -0.046) and raised it under noise (0.070, 0.036 to 0.104).

5.3. Evidence ratio (family S-Z)

GATE-M increased the evidence ratio d_R of Eq. (8) under all three corruptions (Wilcoxon signed-rank test, Holm over three): by 0.0090 (95 % interval 0.0078 to 0.0102; $p_{\mathrm{Holm}} = 1.9 \times 10^{-39}$) under JPEG, by 0.0038 (0.0030 to 0.0047; 8.5×10^{-16}) under noise and by 0.0078 (0.0069 to 0.0087; 2.9×10^{-43}) under splice. This is not a result about verification accuracy. The ratio rises when tokens with extreme per-token evidence are removed for any reason, and the plan states that the random gates do not separate removing corrupted tokens from removing heavy-tailed ones. Three descriptive contrasts bear on that caveat (paired bootstrap 95 % intervals, no p-value). The geometry-matched random gate did not raise d_R: GATE-M minus that gate was 0.0091 (0.0078 to 0.0104), 0.0040 (0.0030 to 0.0049) and 0.0081 (0.0071 to 0.0092), close to GATE-M minus none. The plain-B top-k gate raised d_R almost as much as GATE-M (GATE-M minus it: 0.0008, 0.0001 to 0.0015; 0.0003, -0.0001 to 0.0007; 0.0005, -0.0001 to 0.0011), and the s_3 top-k gate less (GATE-M minus it: 0.0051, 0.0035 to 0.0067; 0.0033, 0.0022 to 0.0044; 0.0045, 0.0033 to 0.0057). On the size-scaled z_R (sensitivity endpoint, unadjusted Wilcoxon p) the change was 0.056 (0.025 to 0.087; $p = 2.5 \times 10^{-4}$) under JPEG, 0.011 (-0.012 to 0.034; $p = 0.35$) under noise and 0.042 (0.017 to 0.066; $p = 0.0012$) under splice, and on the clean test images GATE-M raised z_R by 0.058 (0.032 to 0.085).

5.4. Pixel denoising as a defence (family S-DN)

Family S-DN compares GATE-M with no gate on a copy of the corrupted image cleaned by the pixel denoiser (exact McNemar, Holm over JPEG and noise). Under noise the denoised copy gave higher verification accuracy than GATE-M: 0.852 against 0.676 ($b = 34$, $c = 122$; GATE-M minus denoised -0.176, 95 % interval -0.224 to -0.130; $p_{\text{Holm}} = 1.5 \times 10^{-12}$), against 0.880 on the clean images. Under JPEG no difference was detected: 0.844 against 0.836 ($b = 38$, $c = 42$; -0.008, -0.042 to 0.026; $p_{\text{Holm}} = 0.74$). At this setting the denoiser lowers the quality of JPEG images (Section 4.6), so the JPEG row compares GATE-M with a defence that harms the image. The denoised copy against no gate was 0.176 (0.130 to 0.222) under noise and 0.006 (-0.030 to 0.040) under JPEG. Splice has no denoised copy.

5.5. Calibration of the gate and the clean control (family S-C)

On the 500 clean test pairs the realised flagged share was 0.0456 at $\alpha = 0.05$, with an item-bootstrap 95 % interval of 0.0442 to 0.0470 (Table 8). The calibration statistic was -3.59, against the violation bound of 2.13, so there is no evidence of violation; at $\alpha = 0.01$ and 0.10 the shares were 0.0064 and 0.0956, also with no evidence of violation. In 2 000 random re-splits of calibration and test items at $\alpha = 0.05$ (exploratory, descriptive) the mean of $\hat{F} - \alpha$ was -0.0031, with 2.5 and 97.5 percentiles of -0.0055 and -0.0006. The statement checked is the one of Proposition 1, the expected flagged share of a clean pair's tokens; a pass is "no evidence of violation" and not a validation.

Table 9 gives, with equal prominence, what the guarantee does not cover: the false-alarm proportion on the unchanged tokens of corrupted items, the hit rate on changed tokens and the precision among flagged tokens, at $\alpha = 0.05$. On the unchanged tokens of the realistic corruptions the false-alarm proportion was 0.036 to 0.044; under swap it was 0.089, above α. Precision must be read against the share of changed tokens: under JPEG and noise more than nine in ten tokens change, so a precision of 0.976 and 0.922 is close to that base rate (0.961 and 0.928); under noise the hit rate (0.033) was below the false-alarm proportion (0.036).

The clean no-gate accuracy on the test fold was 0.880 (Wilson 95 % interval 0.849 to 0.906), against 0.843 (0.798 to 0.880) in the pilot; the intervals overlap, so the pre-stated pipeline audit was not triggered. The same check gave 0.880 (0.838 to 0.912) on the calibration fold and 0.860 (0.779 to 0.915) on the headroom fold. In the clean control (family S-C,

exact McNemar, one hypothesis) GATE-M on clean images gave 0.882 against 0.880 without a gate ($b = 2$, $c = 1$; net 0.002, 95 % interval −0.004 to 0.010; $p = 1.00$): no change was detected.

5.6. Positive control (family PS)

The positive control repeats the primary test on the two sanity corruptions, with Holm correction over two (Table 10). Neither hypothesis was rejected, so the positive control was *not met*. By the plan this reads "pipeline defect suspected, audit triggered"; nothing was tuned. Both sanity corruptions lowered the clean accuracy of the test fold from 0.880 to 0.870 without a gate. The audit of the analysis plan (SAP, A2.3) was run on 29 October 2026, after the outcome of PS was known. It had four parts: the unit tests of the code that produced the run (584 passed, 5 skipped, none failed); the layout-null check and the tokenisation self-check recorded by the run itself; a code read of the path that produces PS for errors of sign, indexing and layout; and an exact recomputation of every PS quantity from the stored arrays of all 500 items, which found no mismatch. The result is: positive control not met; audit triggered (SAP A2.3); no defect found. The audit changed no number, rule, threshold or label. The realised discordance, which the plan requires to be printed with this result, was $n_d = 4$ for patch and $n_d = 13$ for swap (Table 10). The finding of no defect is limited to the classes of error that the plan names and to the path from the stored arrays onwards: a recomputation from the stored arrays cannot see an error upstream of them, which the run's own records (a maximum absolute difference of 0.0 in the layout-null check and a tokenisation self-check equal to the reference) argue against but do not exclude. The audit is not read as a validation of the pipeline. The positive control says nothing about where the gate works. A post hoc, descriptive account of how much room for a change in the decision these two corruptions leave is in Section 6.1; it enters no test.

5.7. Detection against cheap rivals (families S-D and S-D2)

This family was analysed before the main test, on the 200 detection items, and it is a measurement of the within-modality score q; it enters no claim about the primary result. Intervals are 95 % percentile bootstrap intervals over items for the per-item AUROC against changed-code labels; items with too few corrupted or clean tokens are excluded and counted (3 for JPEG, 2 for noise). On the sanity corruptions q reaches 0.961 (0.959 to 0.963) for a replaced patch and 0.955 (0.952 to 0.957) for swapped codes, so the chance

bar of the positive control was passed; plain-B is slightly higher, 0.964 and 0.960, with q minus plain-B equal to -0.0033 (-0.0042 to -0.0026) and -0.0053 (-0.0058 to -0.0047). The non-finetuned TbV score reaches 0.854 and 0.859 and the neighbour codebook distance 0.786 and 0.741 on the same corruptions; a check for copying of the released output layer suspected no item. On JPEG quality 10 q reaches 0.561 (0.547 to 0.575, 197 items) and on Gaussian noise 0.10 it reaches 0.503 (0.493 to 0.513, 198 items). The non-finetuned TbV score reaches 0.581 (0.569 to 0.593) on JPEG and 0.607 (0.597 to 0.618) on noise; on noise the marginal intervals of q and TbV do not overlap, on JPEG they do, and no paired test between them was run. The best cheap rival is the pixel variance of a token's patch with its sign flipped, 0.756 (0.743 to 0.768) on JPEG and 0.768 (0.760 to 0.776) on noise. The rival and its sign were chosen on the data among about nine candidates, which biases the contrast against q. The paired difference of q minus the rival is -0.195 (-0.214 to -0.175) on JPEG and -0.265 (-0.278 to -0.253) on noise, with a Holm-adjusted p of 0.0004 on each, the resolution floor of 10 000 resamples. The result is the same when the rival is re-selected in every resample and when the 21 items analysed before the plan's second addendum are removed (179 items). The preregistered falsifier for the score was therefore met on both corruptions, and the paper does not say that the score detects corrupted codes better than cheap methods. That a texture score wins against this label is read as the label-circularity concern of Section 6.3 and not as evidence that texture explains the corruption effect. Figure 2 shows the full detection map.

The out-of-sample family S-D2 repeats the contrast on the test items at the operating points, with the rival frozen on the complete detection set (patch texture, sign flipped) and Holm correction over two. Under JPEG quality 10 q reached 0.561 against 0.759 for the rival (492 items with enough tokens of both kinds; q minus rival -0.198, 95 % interval -0.211 to -0.186; $p_{\text{Holm}} = 0.0004$), and under noise 0.509 against 0.767 (497 items; -0.258, -0.267 to -0.250; $p_{\text{Holm}} = 0.0004$), again at the resolution floor. S-D2 has no splice member, so the AUROC of q under splice was not measured; its flag rates at $\alpha = 0.05$ are in Table 9.

5.8. Decomposition of the accuracy lost to corruption (E-dec, descriptive)

E-dec splits the accuracy lost to corruption, $T = A_{\text{clean}} - A_{\text{none}}$, into the residual loss under the oracle gate $U = A_{\text{clean}} - A_{\text{chg}}$, the realised recovery by GATE-M $R = A_{\text{GATE-M}} - A_{\text{none}}$ and the gap to the oracle gate $V =$

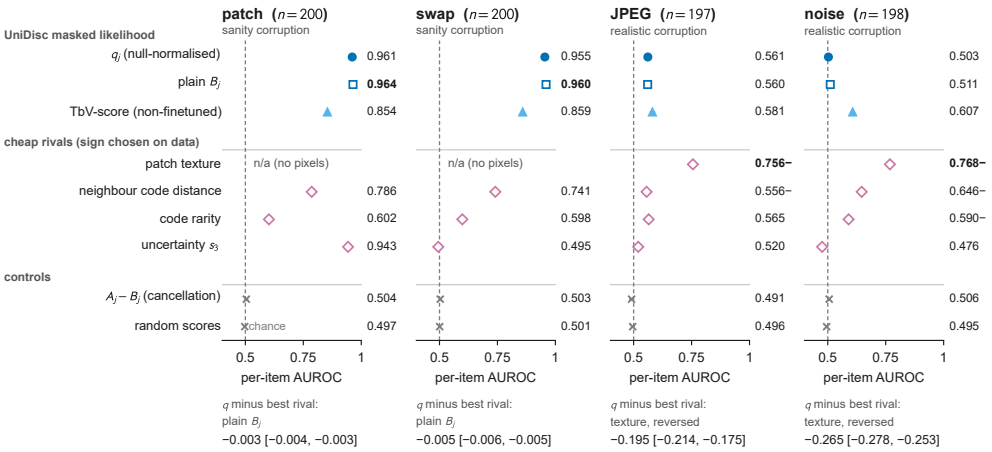


Figure 2: Detection map: per-item AUROC of each score against the changed-code labels, by corruption, on the 200 detection items (secondary family S-D; not a test of the gate effect). Markers are the mean over items; bars are 95 % percentile bootstrap intervals over items (10 000 resamples), narrower than the marker for most rows; numbers at the right are the means (bold: highest in the panel). n: items with enough corrupted and clean tokens (200 patch and swap, 197 JPEG, 198 noise); texture needs a pixel image and is not applicable to patch and swap. Blue: the within-modality masked likelihood of UniDisc (q_j and plain B_j); sky blue: TbV score, non-finetuned. Reddish purple, hollow: cheap rivals, each in the better of its two orientations chosen on these data (a trailing minus marks the reversed one), so rival values are optimistic. Grey: controls ($A_j - B_j$ and random scores); dashed line: chance, 0.5. Under each panel: q minus the best rival with its 95 % interval. No claim is made that q detects better than the cheap rivals; the texture result is a label-circularity concern, not an account of the corruption effect. Source: `runs/P2-GATE-M1/analysis.json`, keys `table.*`, `falsifier.per_corruption` and `paired_q_vs_best_cheap`; script `paper2/figures/src/fig_detection_map.py`.

$A_{\text{chg}} - A_{\text{GATE-M}}$, so that $T = U + R + V$ item by item (Table 11). The oracle is the changed-code gate. It is descriptive: no test, no p-value, and no part is an account of the primary result. Under JPEG the gap to the oracle gate was negative with an interval that excludes zero ($V = -0.078$, -0.116 to -0.040), which by the plan voids any detection reading of the JPEG cell. Under noise the oracle gate recovered part of the loss ($V = 0.070$, 0.036 to 0.106, of $T = 0.204$). Under splice the loss itself was not distinguishable from zero ($T = 0.004$, -0.006 to 0.014). On the evidence ratio d_R the same split gave $T = 0.061$ (0.048 to 0.075), $R = 0.0090$ (0.0078 to 0.0101), $V = 0.073$ (0.035 to 0.115) under JPEG; $T = 0.119$ (0.099 to 0.141), $R = 0.0038$ (0.0030 to 0.0047), $V = 0.083$ (0.042 to 0.116) under noise; and $T = 0.020$ (0.017 to 0.023), $R = 0.0078$ (0.0069 to 0.0088), $V = 0.018$ (0.014 to 0.022) under splice. On the positive-control cells (accuracy scale) T was 0.010 (0.002 to 0.020) for patch and 0.010 (-0.002 to 0.022) for swap, with $R = 0.000$ (-0.008 to 0.008) and 0.006 (-0.008 to 0.020). The parts are relative to the changed-code label, which marks encoder instability and is not an evidence oracle: U contains a composition effect, because the removed positions carry different evidence from the rest, and V mixes false alarms, misses, trimming of the variance and the shape of the gate.

5.9. Sign agreement of the independent-token calculation (E-pred, descriptive)

E-pred compares, cell by cell, the sign s_c of the plug-in signal-to-noise change of Eq. (11), fixed on a separate fit chunk and committed before the first test item, with the observed sign o_c of the net accuracy difference $b - c$ (Table 12). It has no test, no p-value and no level. On the primary cells the sign of the plug-in signal-to-noise change agreed with the sign of the net accuracy difference in 2 of 2 cells (95 % Clopper–Pearson interval 0.158 to 1.000); the noise cell had no observed sign ($b = c = 0$) and was excluded. The rival rule "always +" agreed in 0 of 2 cells and the rule "− for primary cells, + for positive-control cells" in 2 of 2. On the positive-control cells, whose outcome was foregone by design, the signs agreed in 0 of 1 (0.000 to 0.975); the patch cell had no observed sign. No cell had an observed interval that excludes zero, so the refutation rule of the plan (a sign that disagrees with such a cell) was not met in any cell. The analysis file also records the comparison on the evidence ratio d_R, which GATE-M raised in all three primary cells with intervals that exclude zero (Section 5.3) while the plug-in sign was negative; the plan fixes the observed sign on correctness, and d_R rises when extreme tokens are removed for any reason, so this disagreement is

reported and not read either way. The plug-in estimates on the test fold were, for JPEG, noise and splice: changed share ρ 0.961, 0.928, 0.227; clean-token mean μ 0.213, 0.182, 0.202 and variance σ^2 0.501, 0.431, 0.510; changed-token mean μ_c 0.169, 0.099, 0.132 and variance σ_c^2 0.556, 0.456, 0.492; false-alarm rate $\hat{\phi}_0$ 0.037, 0.036, 0.044; hit rate $\hat{\phi}_1$ 0.061, 0.033, 0.049.

5.10. Re-infilling the flagged tokens (family S2)

[RESULT (run P2-GATE-S2; pending): status "run, formal", "run, descriptive ($n < 120$)", "partial prefix" or "not run" with the reason. If run: patch and splice, I-GATE against I-RAND4 (kept draw, exact McNemar, Holm over two) with the draw-level and all-four contrasts beside it and I-GATE against none printed beside it; completed n. The claim-bearing cell is splice; a patch-only rejection is a positive control with no realistic cell. No statement for JPEG or noise; the string "G.1" is not used. No sentence relates S2 to the primary result (AB.1).]

5.11. Two-way fusion on hard negatives (family B)

[RESULT (run P2-GATE-FOIL; pending): status "run, formal", "untestable at ceiling" (with the clean accuracy of the better single direction on the fit pairs) or "not run" with the reason. If run: fusion against the better single direction (exact McNemar, one hypothesis), the two margins and the frozen fusion line; text-direction conditioning arms for the three realistic corruptions with intervals (exploratory). B says nothing about corruption and nothing about the primary result (AB.1). If B ran before S2, state the logged order and the inversion risk.]

5.12. Second model (family R)

Family R was not run. It is gatekept by the primary family: it repeats the primary test on a second model only for corruptions with a primary rejection with $b > c$, and there was none.

5.13. Exploratory corruption suite

[RESULT (pending): the exploratory corruption suite (eight cells, intervals only, no p-value in any claim), or "not run" with the reason; never in the abstract or highlights and never pooled with E-pred.]

5.14. Exploratory: hard gate against detector-free combiners, discounting and reliability weighting of the same evidence

This comparison was pre-specified in Addendum A4 (Section 4.11) and is exploratory (Tier 3): intervals only, no p-value, 72 intervals in all and no multiplicity adjustment. It was run once, after the primary analysis was committed, on the same per-token evidence Δ_j, items and decision rule as the primary family, and it cannot explain, qualify or rescue the primary result. Each comparator is contrasted with the hard gate (GM, which is GATE-M) on the same items. The reading of a contrast comes from the exact Clopper–Pearson 95 % interval for the share b/n_d of discordant items won by the comparator: *above* GM if its lower end exceeds one half, *below* if its upper end is under one half, *identical* if $n_d = 0$, and *not distinguishable* otherwise. Beside it are printed the net difference with a percentile bootstrap 95 % interval (descriptive) and the bound used for added value: for $n_d < 10$ the exact 95 % upper bound B on the discordance rate, which bounds the absolute net difference, and for $n_d \geq 10$ the minimum detectable difference (MDD, two-sided 5 %, 80 % power, at the realised spread). Tables 13 and 14 give every cell and every comparator.

On the realistic corruptions no detector-free combiner gave higher verification accuracy than the hard gate. The hard gate gave higher verification accuracy than the median under JPEG (-0.032, 95 % interval -0.052 to -0.014; exact 95 % interval for the discordant split 0.047 to 0.374) and under splice (-0.032, -0.050 to -0.016; 0.001 to 0.273), and higher than the 20 % trimmed mean under JPEG (-0.020, -0.034 to -0.006; 0.018 to 0.428) and splice (-0.026, -0.040 to -0.014; 0.000 to 0.247); all exploratory. The data do not show a difference between the hard gate and the trimmed mean at the gate's flag count under JPEG (-0.002, -0.010 to 0.006; $n_d = 5$; $B = 0.023$), noise (0.000, -0.006 to 0.006; $n_d = 2$; $B = 0.014$) or splice (-0.006, -0.014 to 0.000; $n_d = 3$; $B = 0.017$), nor, under noise, between the hard gate and the median (0.002, -0.016 to 0.020; $n_d = 19$; MDD 0.025) or the 20 % trimmed mean (0.006, -0.006 to 0.018; $n_d = 9$; $B = 0.034$); all exploratory. Because each of these bounds is at most 0.05, each counts, by the pre-stated rule, as a finding against the added value of the detector over a detector-free combiner on that cell; by limit (c) below, the contrast with the trimmed mean at the gate's flag count largely repeats the primary contrast when the gate's score is near chance. Discarding and discounting the gate-flagged tokens were not distinguishable under JPEG ($n_d = 1$, $B = 0.011$) or splice ($n_d = 2$, $B = 0.014$), and made the same decision on all 500 items under noise ($n_d = 0$;

the net difference in accuracy is at most 0.007, exact 95 % bound on the discordance rate). Discarding and reliability weighting of every token were not distinguishable on any realistic corruption ($n_d = 15$, 18 and 12; MDD 0.022, 0.024 and 0.020). Against no gate, no comparator gave higher accuracy on a realistic corruption; the median and the 20 % trimmed mean gave lower accuracy than no gate under JPEG and splice, and on clean pairs too (exact readings; keys `contrasts.*_minus_NONE` of the same file).

On the evidence ratio d_R (descriptive control of the plan) the trimmed mean at the gate's flag count gave a higher d_R than the hard gate on every cell: by 0.033 (95 % interval 0.028 to 0.039) under JPEG, 0.018 (0.014 to 0.022) under noise, 0.031 (0.027 to 0.035) under splice and 0.033 (0.029 to 0.038) on clean pairs, and the 20 % trimmed mean by 0.259 (0.225 to 0.292), 0.195 (0.153 to 0.237), 0.303 (0.270 to 0.334) and 0.324 (0.291 to 0.356); exploratory. Symmetric trimming lowers the spread of the retained evidence mechanically, which is why this control was specified.

On the positive-control cells (Table 14; pipeline check only) reliability weighting of every token gave higher verification accuracy than the hard gate under swap (0.016, 95 % interval 0.004 to 0.030; exact 95 % interval for the discordant split 0.516 to 0.979; exploratory) and than no gate (0.022, 0.006 to 0.038; 0.566 to 0.962); this concerns discarding against weighting on a sanity corruption and is not a finding about the realistic corruptions.

5.15. Coherence of the denoiser across mask patterns

On 100 clean calibration items, with four image-token pairs per item and two visible contexts per pair (800 gaps), the median absolute order gap of the joint log-likelihood was 0.071 nats. The same computation repeated in a different batch order on 25 items gave a 95th percentile of 0.000 for the 200 absolute repeat differences, so at least 95 percent of the repeats reproduced the original value exactly. Neither pre-stated reading applies. The gap exceeds the numerical noise floor, so the denoiser is not "coherent within numerical noise"; it is below the clean mean per-token paired evidence $\mu = 0.199$ measured on the 500 clean test items, so it is not "incoherent at the evidence scale", and Eq. (6) is not declared inapplicable. By pair type (descriptive, not a pre-stated reading) the median gap was 0.253 for horizontally adjacent tokens, 0.246 for vertically adjacent ones and 0.032 for distant ones; the two adjacent medians exceed μ.

5.16. Status of every family and cost

Table 15 gives the status of every family; a status is set by time and by the gatekeeping rules, never by an outcome.

[RESULT: network passes per item by stage, median time per item and peak GPU memory, from the run configuration and telemetry files of runs P2-GATE-M2 to M5 (not yet extracted). The operational stops of the main run are in the deviations box (Appendix A.4).]

6. Discussion and limitations

6.1. Findings

The preregistered test excludes an improvement or a loss of more than five percentage points in image-direction verification accuracy from gating under JPEG quality 10, Gaussian noise with $\sigma = 0.10$ and a 128-pixel splice, for this model and these severities; it does not show that gating has no effect.

The primary table shows how narrow the measured change is: in 500 items the gate changed the decision once under JPEG, never under noise and four times under splice (Table 5). The exclusion of ± 0.05 is a statement about these discordances at the threshold calibrated for $\alpha = 0.05$, at one severity per corruption, in one model and on one data set. It says nothing about a gate that flags a larger share of tokens, about stronger corruptions, or about another model or task, none of which was tested. Section 3.7 gives, under independent tokens, the condition on the hit and false-alarm rates under which discarding flagged tokens raises the signal-to-noise ratio of an additive evidence sum (Eq. (11)). The condition states what a gate would have to achieve; it is an approximation for dependent image tokens, the test was not designed to check it, and we do not use it to account for the result.

The remaining findings are separate measurements, each under its own family label, and none of them is an account of the primary result. For each we give the observation, a hypothesis for it, and what would falsify the hypothesis.

The clean false-alarm statement held; the flags on corrupted images are not covered by it. On held-out clean pairs the realised flagged share was 0.0456 at $\alpha = 0.05$ (95 % interval 0.0442 to 0.0470), with no evidence of violation at any of the three levels (Section 5.5). On the unchanged tokens of corrupted images the false-alarm proportion was 0.036 to 0.044 for the realistic corruptions and

0.089 under swap, above α. The conformal statement covers clean pairs only, so these proportions lie outside it; the audit of the positive control did not treat the 0.089 as a defect for that reason. Hypothesis: a token next to a replaced code is scored with damaged context and is flagged more often, as Section 3.4 anticipates. It would be falsified if, under swap, the false-alarm proportion of an unchanged token did not fall with its distance to the nearest replaced code; this was not measured.

The gate's score was near chance at finding the codes that JPEG and noise change. On the detection items and again on the test items (families S-D and S-D2) the per-item AUROC of q was 0.561 under JPEG and 0.503 and 0.509 under noise, below a texture score whose sign was chosen on the data; under noise the hit rate of the gate (0.033) was below its false-alarm proportion on unchanged tokens (0.036). Hypothesis, the label-circularity concern of Section 6.3: JPEG quality 10 and noise change more than nine in ten codes of an image (Table 9), so a changed code marks where the tokenizer is unstable more than where the image is damaged, and a texture score tracks that instability. It would be falsified if q stayed at chance against a damage label defined in the pixel domain, for example the reconstruction error of each token's patch, under which texture had no advantage; this was not measured.

A pixel denoiser restored most of the accuracy lost to noise. Under noise the denoised copy reached 0.852 against 0.676 without a gate and 0.880 on clean images, and it gave higher verification accuracy than GATE-M (family S-DN, Section 5.4); under JPEG no difference was detected, with a denoiser setting that lowers JPEG image quality. Hypothesis: noise of this strength is largely removed in the pixel domain before tokenisation, so the tokens of the denoised image are close to the clean ones. It would be falsified if the share of codes that differ from the clean tokenisation were not lower for the denoised noisy images than for the noisy ones; this was not measured. The denoiser is a measured comparator, not part of the gate, and one setting was used.

Withholding every changed code lowered accuracy under JPEG. The changed-code gate, which keeps only the codes that the corruption left unchanged, lowered accuracy under JPEG by 0.084 and raised it under noise by 0.070 (Table 7); in E-dec the gap to this oracle was negative under JPEG with an interval that excludes zero, which by the plan voids any detection reading of that cell (Section 5.8). Hypothesis: under JPEG this gate decides on about four percent of the tokens of an image, and the sum over so few tokens is noisier than the sum over all of them, which is the composition effect named in the plan. It would be falsified if keeping a random set of the

same small size, instead of the unchanged codes, did not lower accuracy by a similar amount; this was not measured.

Exploratory: no detector-free combiner was above the hard gate on a realistic corruption. In the exploratory comparison (Section 5.14; Tier 3, intervals only), the hard gate gave higher verification accuracy than the median and the 20 % trimmed mean under JPEG and splice; the data do not show a difference between the hard gate and the trimmed mean at the gate's flag count on any realistic corruption, nor between the hard gate and the median or the 20 % trimmed mean under noise, with bounds at most 0.05, which counts by the pre-stated rule as a finding against the added value of the detector on those cells. Discarding the flagged tokens and discounting or weighting them were not distinguishable on any realistic corruption. This is a property of these combiners on these cells, measured on the same evidence as the primary test, and not an account of the primary result; when the gate's score is near chance, the comparison with the trimmed mean at the gate's flag count largely repeats the primary contrast (limit (c) of Table 13).

The positive control was not met; the audit it triggered found no defect. On the two sanity corruptions the gate did not change accuracy detectably (Section 5.6): positive control not met; audit triggered (SAP A2.3); no defect found, with $n_d = 4$ (patch) and $n_d = 13$ (swap) discordant items. The audit covers the classes of error that the plan names and does not validate the pipeline; Section 5.6 states what it cannot see. Nothing in the positive control is used in this paper to account for the primary result.

Post hoc and descriptive: the decision headroom of the positive control. This paragraph was written after the outcome of PS was known, enters no test, and describes the task and the design of PS, not the gate. Without a gate, the sanity corruptions changed the decision on 5 of the 500 test items under patch (all five from right to wrong) and on 9 under swap (7 from right to wrong, 2 from wrong to right), that is on 0.010 and 0.018 of the items. With no losses, the exact two-sided McNemar test at the first Holm step (level 0.025) needs at least 7 wins; a gate that returned every item to its decision on the clean image would have given $b = 5$, $c = 0$, $p = 0.0625$ under patch and $b = 7$, $c = 2$, $p = 0.18$ under swap, and would not have reached that step. The changed-code arm, which uses the true set of replaced codes, did not either ($b = 4$, $c = 2$, $p = 0.69$ under patch; $b = 7$, $c = 4$, $p = 0.55$ under swap), and GATE-M made the same decision as that arm on 498 of the 500 items in each cell. The discordance between GATE-M and no gate was 0.008 under patch and 0.026 under swap, against the 0.05 to 0.20 that the power

calculation of the plan assumed (AD.4, where it is marked as an assumption). At that discordance this test could not have rejected for a perfect gate, so the positive control, as designed, cannot separate a working pipeline from a broken one on these two corruptions; this is a property of the task, where replacing 64 or 154 codes rarely turns a correct decision into a wrong one, and of the design, which assumed more discordance than occurred, and not a statement about the method. The description is not used to account for the primary result, and the positive control is not read as met on this ground.

6.2. Where the work sits in information fusion

What is fused. The 1 024 image tokens of one image are 1 024 sources. Each reports a log-likelihood ratio between two candidate captions, and the decision sums them; a second, text-direction statistic can be fused with that sum at score level (family B). *At what level.* At the input: the gate acts on the sources before the evidence sum, not on the output decision, which is why the paper speaks of input-level source validation. *Where reliability enters.* Only through a zero-one weight in the sum (GATE-M) or through replacement of the flagged values by a model draw (GATE-I). The weight is computed from within-modality evidence alone, and its threshold is set on clean data at a fixed false-alarm level, as in CFAR detection. *What is not claimed.* The paper claims no new fusion rule. The additive rule is optimal only for sources that are independent given the hypothesis, which image tokens are not, and the hard weight is not claimed to be the best way to use a reliability estimate; graded weights and robust aggregation are the standard alternatives, and Section 4.11 states the exploratory comparison with them.

The calibrated gate of Zhou and Lu [30] and the gate studied here answer different questions. As described in the abstract of Zhou and Lu, theirs decides whether to keep a default prediction or override it with a challenger's for one test instance; the gate here decides which input tokens to withhold before two directions of image-text evidence are combined. The two could be stacked, but this is not tested. Evidential fusion [13, 14] and conformal prediction on a fusion network [47] are other uses of reliability and of distribution-free calibration in this journal that we found; as far as their abstracts describe, none scores input tokens with the likelihood of a generative model. **[AUTHOR: after the results are frozen, give the abstract and Figure 1 caption to a reader with no project context and ask what is fused (open check on the scope-fit question); settle the scope-fit question with the supervising professor before submission.]**

6.3. Limitations

Except for the two items on the positive control, the limitations below are fixed by the design and do not depend on the outcome.

- *One model, one data set.* All measurements of the primary family use UniDisc 1.4B and COCO val2017 images as tokenised by one tokenizer. A second, 4-bit quantised model was to enter only in family R, only if P had a rejection with $b > c$; P had none, so R was not run. No statement is made for other models, resolutions, caption styles or data sets.

- *Contamination is unknown.* Whether the evaluation images were in the training data of the model or of the tokenizer was not established, and it could make clean pairs easier than unseen ones. The same applies, with more force, to the hard-negative set, whose true captions are COCO captions.

- *Narrow guarantee.* The calibration bounds the expected false-alarm proportion on clean, exchangeable test pairs. It gives no bound on detection power, none on the clean tokens of corrupted items, and none under shift. Unlike a bound on the harm of overriding a prediction, such as a calibrated gate like that of Zhou and Lu [30] may give **[AUTHOR: verify against full text via VIT e-gateway]**, it says nothing about the harm or the power of the gate on corrupted inputs. The realised proportion on one finite test set can lie above α because the statement is marginal over the calibration draw.

- *Corruptions and labels.* Corruption labels are relative to the tokenisation of the clean image, so a "clean" token may already carry compression artefacts of the original photograph. Changed codes under JPEG and noise concentrate where the tokenizer is unstable, so the label may track tokenizer instability more than damage: a texture rival that wins against this label is not thereby shown to explain the corruption effect, and the changed-code gate is not an upper bound for these corruptions.

- *Comparators chosen on the data.* The best cheap rival is selected among about nine candidates, with both signs admitted, which biases the detection contrast against the gate. The rival's sign could not be fixed blind (Section 4.15).

- *Pixel denoiser.* One setting was used for all corruptions; it was chosen after one exploratory evaluation on 20 images and harms JPEG images, so the denoising contrast on JPEG favours the gate.

- *Variance not in the intervals.* Intervals cover item sampling. They do not cover variation across mask partitions, corruption draws, jitter seeds or models, because no training seeds exist and only one corruption draw is made per image. Eleven test families are run with no multiplicity control across them.

- *A constructed task.* The primary decision is a two-way caption match on the weaker of the two available statistics, with random negatives (Section 4.8). It measures how an image-direction statistic tolerates damage and says nothing about what a gate adds to a verifier that uses the stronger text direction; only family B addresses the two-direction setting, and it may be untestable at the ceiling or not run.

- *A hard gate and dependent tokens.* The gate is binary, and the evidence sum and the signal-to-noise analysis assume independent tokens, which neighbouring image tokens are not. Graded and robust alternatives are examined only in an exploratory comparison (Section 4.11), which is not a test and is subject to its own limits.

- *Time input and checkpoint.* The scorer calls the denoiser without a time input; that this is the intended path was checked for the released configuration and checkpoint only, the training weight of whole-modality masks is unknown, and the key check does not cover the image tokeniser (Section 4.1, Appendix B).

- *Scope of the preregistered run.* Families S2 (re-infilling), B (two-way fusion) and R (second model) each have a status of run or not run, decided by time and by their gatekeeping rule and never by an outcome, and the paper reports that status. Caption corruption, the backbone authors' own re-infill recipe and the gating of caption tokens are not tested as families.

- *Positive control not met.* The preregistered positive control on the sanity corruptions was not met, which triggered a pipeline audit (Section 5.6); the audit found no defect. It was run after the outcome was known and covers the classes of error that the plan names. A recomputation

from the stored arrays cannot see an error upstream of them, and the audit does not validate the pipeline. The conformal statement covers clean pairs only; on the unchanged tokens of swapped images the flag proportion was 0.089, above α, which the audit recorded and did not count as a defect.

- *Power assumption of the positive control.* The plan's power calculation for the positive control assumed a discordance of 0.05 to 0.20 between the gated and the ungated decision. The realised values were 0.008 and 0.026 ($n_d = 4$ and 13 of 500), and at that headroom the preregistered test could not have rejected even for a gate that restored every clean decision. The positive control therefore cannot separate a working pipeline from a broken one on these two corruptions. This is a property of the design and of the task, not of the method, and it is not used to account for the primary result.

- *Hardware and missing data.* The GPU is a single laptop device that failed in earlier pilots. A crash leaves a prefix of the item list and the missing-item rule applies; a fold under 240 items is reported as incomplete.

- *Narrow novelty.* Detecting and remasking corrupted input tokens, cross-modal remasking, risk-controlled thresholds for vision-language errors, including gates that decide between whole predictions [30], and masked-likelihood scores each have precedents (Section 2). The claim is a measurement of their composition on one model, and it holds only as far as the measured results go.

- *Concurrent work.* A search for concurrent work was last run on 28 October 2026; unpublished or under-review work is not visible, and the closest paper in this journal [30] was compared from its abstract and preview only. **[AUTHOR: repeat the search once before submission, and read the full text of Zhou and Lu.]**

7. Conclusion

We measured, in a preregistered test on one pretrained unified masked diffusion model (UniDisc, 1.4B parameters) and COCO image-caption pairs, whether withholding the image tokens that the model's own masked likelihood

flags, with a threshold set by conformal risk control on clean pairs, changes image-direction verification under image corruption. The test excludes an improvement or a loss of more than five percentage points in verification accuracy from gating under JPEG quality 10, Gaussian noise with $\sigma = 0.10$ and a 128-pixel splice (90 % intervals of the net change between -0.010 and 0.002; $n = 500$); it does not show that gating has no effect. On held-out clean pairs the realised flagged share was 0.046 against a target of 0.05, with no evidence of violation; separately, in the secondary detection family, the gate's score was near chance at finding the codes that JPEG and noise change (AUROC 0.561 and 0.503). These are measurements on one model, one data set, one severity per corruption and the weaker of two verification statistics, with a guarantee that covers clean pairs only; re-infilling and two-way fusion are reported as run or not run **[RESULT: S2 and B status]**. **[AUTHOR: confirm the future-work sentence.]** The next measurements are a second backbone, a damage label that does not depend on which codes the tokenizer changes, and graded reliability weights inside the evidence sum, each preregistered in the same way.

CRediT authorship contribution statement

Shlok Kumar Goenka: **[AUTHOR: choose from the 14 CRediT roles (conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, writing – review and editing, visualization, ...)]**. **[AUTHOR: supervising professor]: [AUTHOR: roles (for example supervision, project administration, writing – review and editing)]**.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[AUTHOR (Shlok): exact wording of this declaration. The Elsevier template sentence is: "During the preparation of this work the author(s) used [NAME OF TOOL/SERVICE] to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article." Name each tool, its version and its purpose, and the extent of review. Decision still open for Shlok: whether any AI-assisted step of the research process (code, analysis) must

also appear in the methods under the journal's policy; the project rule keeps every AI mention in this section.]

Declaration of competing interest

[AUTHOR: complete the Elsevier declarations tool and state "The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper" only if true.]

Funding

[AUTHOR: funding sentence. If there is none, the fixed Elsevier sentence is "This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors."]

Acknowledgements

[AUTHOR: acknowledgements, if any.]

Data availability

[AUTHOR: repository and DOI. Planned content: the code at commit `164a18a5` **(**`exp_p2/`**,** `tests/exp_p2/`**), the fold file (SHA-256** `450edb543779de7ee5e6a52422fd6507ef52f15ba3b956a8b05c2e561620f311`**), the preregistration card, the analysis plan (SHA-256** `b4e0bf575dfc2524c0f9d3d5c6b5f52e3185088a94ced8c75f092104ed396c47`**), per-item scores and predictions of every run directory cited, the analysis and figure scripts, and the licence. The COCO images and captions and the UniDisc and tokenizer weights are third-party and are cited, not redistributed. Do not write "available on request".]**

Appendix A. Preregistration record

Appendix A.1. Pins

Table A.16 lists the pinned artefacts. Hashes of the card are SHA-256 of the committed bytes (LF convention) unless marked; the worktree is CRLF for some files, so a raw hash can differ without a content change. Twelve-hex prefixes stand for hashes that are given in full in the pin file of the repository.

Appendix A.2. Audit and sign-off record

The analysis plan was signed off by the statistician with changes. The red team signed Addenda A2 and A3 with conditions, and signed Addendum A4 with eleven conditions that revision r1 answers. The code was audited in several rounds by a separate auditor, who found defects that were fixed; the final verdict is a pass at the final audited commit, and a delta audit covers the later change of two command-line flags. The hard-negative id mapping and leak check were signed off by a data steward. No signature says that the method works, and no reviewer read a GPU outcome. The pipeline audit that the positive control triggered (SAP A2.3) was carried out by a separate auditor on 29 October 2026, after the outcome of PS was known; its verdict is no defect found, and its record is `research/paper2/prereg/A2.3-AUDIT-PS-2026-10-29.md` (data: `runs/P2-A23-AUDIT/a23_ps_audit.json`). Its descriptive account of the decision headroom is post hoc and is reported as such in Section 6.1.

Appendix A.3. Families and wording rules

Table A.17 lists the eleven test families and the three descriptive analyses. No secondary family can rescue a null result in P. A null in P is reported with its label and with no causal clause drawn from PS, S2, B, R, E-dec, E-pred, the suite or the exploratory comparison of Section 4.11, in the title, abstract, highlights or conclusion.

Appendix A.4. Deviations and disclosures known at the time of writing

1. The sample size of S2 and B was set to 300 instead of 150 before any outcome, conditional on the first author's dated confirmation; S2 was restricted to patch and splice after a 21-item dry run.
2. The first stage of the detection measurement stopped and was resumed at item 106 of 200.
3. Three checks of the second-model adapter are applied outside the module that holds the others; the adapter checks are executed by the implementer's code and reviewed, not executed, by the auditor, so the adapter is not described as independently verified.
4. The sensitivity analyses of S2 and B read "rejection" as the favourable-direction rejection at the adjusted level; this can over-flag sensitivity to n and never under-flags it, and it enters no claim.
5. The firewall for the S2 smoke test is an honour rule.

6. The swap and patch parameters are stored in one frozen file that stands for two files named in the plan.
7. The chance bar for the sanity corruptions is called blind only on the lead's assertion that no detection-pilot output from item 26 onward was read, which nobody else can verify; the first author's own written co-attestation that he had not looked at any of those outputs is on file, dated before the analysis was run.
8. Where the card forbids any claim from a rejection with $c > b$ and the analysis plan prescribes the sentence "reduced verification accuracy", the plan governs: no claim in favour of the gate is made from such a rejection, and no title carries it.
9. The exploratory comparison of Section 4.11 was pre-specified in Addendum A4, whose draft and revision were committed before the headroom decision was written; the addendum's author states that no value of the headroom or primary stage was read when it was written, which is an assertion and not an attestation. Its code is committed only with a dated pin row.
10. Family B may run before S2 if a fixed time inequality holds; if a stop occurs while B runs, the order reverts, and B complete with S2 partial is then possible.
11. Whether a fallback applies if P is null, and the wording rule that governs the framing of the paper in that case, was **[AUTHOR: confirm the wording. Record (decisions D0076-D0077, CONFIRM-EVENING section 12): on 29 October 2026, about 00:10 IST, the first author delegated the choice; the stricter reading of the outcome book was adopted before any value of the main test was read, so the framing "when does input gating help" stays out of the title, abstract, highlights, conclusion and the opening of the discussion, and the submission condition is unchanged with a status-only definition of "informative". Name who adopted it.]**
12. The calibration stage (300 items) stopped twice on the slow-item rule of the guard, with 106 and then 114 rows complete: an item took 3.0 s against a limit of twice a median of 1.5 s, and later 4.4 s against twice 1.8 s. Neither stop was a crash; the second coincided with host-side load while the GPU waited. On the first author's dated answers the stage was relaunched, first unchanged and then with a limit of three times the median item time and a floor of 2 s, and the same limit applies to every

later stage except the second-model stages (row A-2). Guard settings decide only whether a run continues; the rows already stored were kept unchanged.

13. The main test stage (500 items) stopped once at 334 of 500 on the default limit of 12 chunks of the stage runner (reason "max chunks reached"; not a guard stop and with no crash event) and was relaunched with a limit of 30 chunks, resuming at row 334 with the same item list and the same flags. The preregistered launch command did not print this setting; it is an operational setting and changes no output.
14. During the coherence stage (100 items) the operating system restarted the laptop to install driver updates. The stage's process ended after one of 100 items. The system log records an update-initiated restart, with no kernel-power, bug-check, hardware-error or GPU-driver event, and the GPU driver version was unchanged, so this is not counted as a crash or a guard stop. The stage was continued afterwards on the first author's instruction and completed with 100 of 100 items; automatic updates were paused afterwards.
15. The family order at the end of the schedule was FOIL before S2, which the plan allows if a fixed time inequality holds at the end of the coherence stage (A4.3.2); it held. A separate auditor read the code of both stages (lines 1 to 149 of the FOIL runner among others) and confirmed that no FOIL output can reach an S2 input. FOIL reads no output of the calibration, main-test, coherence, S2 or E-pred stages; it reads, and may add, layout-only null files in a shared cache directory that S2 never reads. The scored path of FOIL uses no random numbers: its only generators are local ones in the start-up checks and the bootstrap seed child used in its analysis. This is a reading of code and not a run, and the rule that the analysis of FOIL waits until the S2 table exists is procedural and not enforced by code.
16. No second-model stage was run. The set P^+ of corruptions with a primary rejection and $b > c$ is empty, and by the plan (AD.7) family R is run only for P^+.
17. The smoke test of S2 stopped twice at the tokenisation self-check, before any item, because 0.9854 and then 0.9844 of the image tokens matched the reference where the rule requires 0.99 (the main-test stage gave 1.0). The cause was process-to-process nondeterminism of the autotuned TF32 convolutions in the image tokeniser: the same image gave 0.998 in one fresh process and 0.985 in another, with no S2 code

involved. The self-check was changed to run with cuDNN autotuning off and deterministic algorithms on, with the previous settings restored afterwards; the 0.99 rule, the scoring and the sampler are unchanged (dated row A-5, code commit `8d1bee3a`, passed a delta audit). The check then passed. S2 reads the stored codes of the main-test stage and never tokenises an image afresh, so no S2 output depended on the check. The directories of the two failed smoke attempts are kept.

Appendix A.5. Dated amendments

Row A-1 moved the code pin to the commit named in Table A.16 for every stage launched afterwards; it changed two command-line flags of the stage runner and the guard settings recorded per chunk, and no analysis path, scoring definition, frozen file or guard. Row A-2 appended a slow-item limit of three times the median item time, with a floor of two seconds, to every non-second-model stage, on the first author's dated answer; guard settings decide only whether a run continues and never change an item's output. Row A-5 moved the code pin to commit `8d1bee3a` for the S2 stage and later stages; it changed only the S2 runner, so that its unchanged self-check runs with deterministic convolution algorithms, and added one test (deviation list above).

Appendix A.6. Symbol crosswalk

Table A.18 maps the symbols of the preregistration documents and code to the paper's.

Appendix B. Time input of the released configuration

Section 4.1 states that UniDisc's denoiser has no time input in the released configuration. Table B.19 gives the evidence, read in the authors' code at commit `3ee3e47a9e64bac2fdca59957726f14723371ced` of the UniDisc repository (clean tree) and in the key names of the checkpoint, snapshot `2b344ca856bd108b52a2d022a3463dace131249f`. The statement is about this configuration and checkpoint and not about UniDisc in general.

The loader is non-strict by default. A key check on a model built without weights found 342 of 342 checkpoint tensors matched to model parameters with equal shape and dtype, no missing key, no unexpected key, and no shape mismatch, and the framework's own loader reports empty lists of missing and unexpected keys. Deterministic rotary-embedding buffers are computed at

construction and are not part of any state dictionary. The check covers the denoiser backbone, not the image tokeniser, which is loaded by another path, and it does not show that the checkpoint values are the intended final weights. During training, a whole modality was masked with some probability; which value applied to the released checkpoint is not recoverable from the code.

Appendix C. Diagnostics, cost and hardware

Appendix C.1. Calibration diagnostics and pipeline checks

On the clean test items the realised false-alarm proportion $\hat{F} = n^{-1}\sum_i L_i(\hat{\tau})$ is reported for $\alpha \in \{0.01, 0.05, 0.10\}$, with $\alpha = 0.05$ primary. A violation is flagged when $Z_{\text{cal}} = (\hat{F} - \alpha)/\big(\hat{\sigma}_L\sqrt{1/n_{\text{cal}} + 1/n_{\text{test}}}\big)$ exceeds 2.13 at any level, where $\hat{\sigma}_L$ is the pooled item-level standard deviation of L_i; a pass means "no evidence of violation" and is never worded as validation. The precision among flagged tokens, the false-alarm proportion on the clean tokens of corrupted items and the spatial map of clean false alarms are reported with equal prominence; for $\alpha = 0.05$ on an 8×8 patch image, about 48 false flags are expected against at most 64 true ones, so precision is at most 0.57 even for a perfect detector. For the sanity corruptions the lower bound of the 95 % interval of the per-item AUROC of q and of plain-B must exceed 0.5; a result at chance is a pipeline defect, not a scientific result, and triggers an audit. The clean no-gate accuracy is reported with a Wilson interval against the pilot value of Section 4.13 as a descriptive check. A coherence test on 100 clean calibration items measures whether the denoiser's joint likelihood of two image tokens is independent of the order in which they are revealed; it is read as "coherent within numerical noise" if the median absolute gap lies within the 95th percentile of the gap between repeats of the same computation, and as "incoherent at the evidence scale" if it exceeds the clean mean per-token paired evidence, in which case Eq. (6) is reported as not applicable.

Appendix C.2. Cost in network calls

Cost is counted in network passes (one row of a batch), because a call at this scale is the dominant term. Table C.20 lists the passes per quantity. Gating the image adds K passes per image version and nothing else to the image-direction evidence, because the evidence passes depend on the caption only. With only the image gated, verification of N candidates costs K gate

passes, $2N$ evidence passes (the image-direction half cached per caption), and $S_r n_{\mathrm{draw}}$ more for GATE-I, against $2N$ for the ungated two-way baseline.

Appendix C.3. Hardware and runtime

All runs use one laptop GPU (NVIDIA RTX 5070 Ti Laptop, 12 GB) with 31 GB of system memory and no other hardware. In the pilot one item cost 58 network passes, the median item time was 7.0 s and the peak GPU memory 8.2 GB, so passes are batched four rows at a time. Runs go in short checkpointed chunks under a guard that stops on memory, temperature and power-source events and is resumable; an abort ends the stage and is not retried until reviewed. An earlier detection pilot stopped after 20 complete items when the GPU session failed. For reproducibility: image tokenisation in a fresh process runs with autotuned TF32 convolutions, and two processes can differ by up to about 1.5 % of the 1 024 tokens of an image (self-check agreement 0.998 to 1.0 in earlier stages, 0.984 and 0.985 in the S2 smoke test); the stored codes of the main-test stage are the reference, and S2 reads them and does not tokenise images afresh.

Appendix D. Proof of Proposition 2

Write $m_0 = (1-\rho)m$ and $m_1 = \rho m$ for the numbers of clean and corrupted tokens, $k_0 = 1-\phi_0$ and $k_1 = 1-\phi_1$ for the keep probabilities, and $z_j \in \{0,1\}$ for the label of token j, so that $\mathbb{E}\Delta_j$ is μ or μ_c and $\mathrm{Var}\Delta_j$ is σ^2 or σ_c^2. The exchangeable block covariance of assumption (a) is positive semi-definite if and only if $\gamma_0 \geq -1/(m_0-1)$, $\gamma_1 \geq -1/(m_1-1)$, $\gamma_0, \gamma_1 \leq 1$ and $\gamma_{01}^2\, m_0 m_1 \leq \{1+(m_0-1)\gamma_0\}\{1+(m_1-1)\gamma_1\}$.

Variance.. By assumption (b), I_j is independent of Δ_j, so $\mathrm{Var}(I_j\Delta_j) = k(\sigma_z^2+\mu_z^2) - k^2\mu_z^2 = k\sigma_z^2 + k(1-k)\mu_z^2$ with $k = k_{z_j}$, which are the terms of Eq. (10). For $j \neq k$, again by (b), $\mathrm{Cov}(I_j\Delta_j, I_k\Delta_k) = \mathbb{E}I_j\,\mathbb{E}I_k\,\mathbb{E}[\Delta_j\Delta_k] - \mathbb{E}I_j\,\mathbb{E}I_k\,\mathbb{E}\Delta_j\,\mathbb{E}\Delta_k = k_{z_j}k_{z_k}\,\mathrm{Cov}(\Delta_j,\Delta_k)$. There are $m_0(m_0-1)$ ordered clean pairs with covariance $\gamma_0\sigma^2$, $m_1(m_1-1)$ ordered corrupted pairs with covariance $\gamma_1\sigma_c^2$ and $2m_0m_1$ ordered mixed pairs with covariance $\gamma_{01}\sigma\sigma_c$, which gives Eq. (12).

Condition.. Let V_g be Eq. (12) at (ϕ_0,ϕ_1) and V_0 its value at $\phi_0 = \phi_1 = 0$. With $\mu_c = 0$, $\mathbb{E}S = m_0k_0\mu$ with the gate and $m_0\mu$ without it, so, both being positive, the gate raises the ratio if and only if $k_0^2V_0 > V_g$. Collect $k_0^2V_0 - V_g$ by type of term. The clean terms are $k_0^2 m_0\sigma^2\{1+(m_0-1)\gamma_0\} -$

$m_0 k_0(\sigma^2+\phi_0\mu^2) - m_0(m_0-1)k_0^2\gamma_0\sigma^2 = -m_0 k_0\phi_0(\sigma^2+\mu^2)$, in which the γ_0 terms cancel. The corrupted terms are $m_1\sigma_c^2\{(k_0^2-k_1)+(m_1-1)\gamma_1(k_0^2-k_1^2)\}$, with $k_0^2-k_1 = \phi_1-2\phi_0+\phi_0^2$ and $k_0^2-k_1^2 = (\phi_1-\phi_0)(2-\phi_0-\phi_1)$. The mixed terms are $2m_0m_1\gamma_{01}\sigma\sigma_c\,k_0(k_0-k_1)$, with $k_0-k_1=\phi_1-\phi_0$. Dividing the sum by m gives Eq. (13).

Consequences.. (i) is read off Eq. (13). (ii) At $\phi_1=\phi_0\in(0,1)$ every correlation term vanishes and the left side is $\rho\sigma_c^2(\phi_0^2-\phi_0)<0$, below the positive right side. For $\phi_1\le\phi_0$, let $G=(\rho m-1)\gamma_1$, so that $G\ge -1$ by positive semi-definiteness. The bracket $(\phi_1-2\phi_0+\phi_0^2)+G(\phi_1-\phi_0)(2-\phi_0-\phi_1)$ is linear in G; it equals $\phi_1-2\phi_0+\phi_0^2\le-\phi_0(1-\phi_0)$ at $G=0$, equals $-\phi_1(1-\phi_1)\le 0$ at $G=-1$, and does not increase for $G\ge 0$; hence it is at most 0. If $\gamma_{01}\ge 0$ the mixed term is at most 0 as well, while the right side is positive for $\phi_0\in(0,1)$. (iii) The derivatives of the left side with respect to γ_1 and γ_{01} are $\rho\sigma_c^2(\rho m-1)(\phi_1-\phi_0)(2-\phi_0-\phi_1)$ and $2\rho(1-\rho)m\sigma\sigma_c(1-\phi_0)(\phi_1-\phi_0)$, both positive when $\phi_1>\phi_0$ and $\rho m>1$, and the right side does not depend on γ_1 or γ_{01}. For fixed $\phi_1>\phi_0$ the γ_1 term grows without bound with $(\rho m-1)\gamma_1$, which gives the limit. With $\gamma_{01}=0$, $x=\phi_1-\phi_0$ and $\chi=\rho\sigma_c^2/\{(1-\rho)(\sigma^2+\mu^2)\}$, Eq. (13) reads $Gx^2-x\{1+2G(1-\phi_0)\}+\phi_0(1-\phi_0)(1+1/\chi)<0$, so for $G>0$ the gate raises the ratio if and only if ϕ_1 exceeds $\phi_0+\big[1+2G(1-\phi_0)-\sqrt{\{1+2G(1-\phi_0)\}^2-4G\phi_0(1-\phi_0)(1+1/\chi)}\big]/(2G)$ (and $\phi_1\le 1$); as $G\to 0$ this tends to the boundary $2\phi_0-\phi_0^2+\phi_0(1-\phi_0)/\chi$ of Eq. (11). (iv) For $\gamma_{01}<0$ and $\phi_1<\phi_0$ the mixed term is positive and grows with m, so it can exceed the remaining terms. An instance with a positive definite covariance is $m_0=900$, $m_1=100$, $\gamma_0=\gamma_1=0.05$, $\gamma_{01}=-0.045$, $\mu=0.1$, $\mu_c=0$, $\sigma^2=\sigma_c^2=1$, for which the smallest eigenvalue of the correlation matrix is 0.9500 and the ratio rises from 0.4892 without a gate to 0.5080 with the gate $(\phi_0,\phi_1)=(0.3,0)$; the instance is an illustration of the algebra and not an estimate from the model. (v) m enters Eq. (13) through $\rho m-1$ and m whenever $\gamma_1\neq 0$ or $\gamma_{01}\neq 0$. □

References

[1] J. Austin, D. D. Johnson, J. Ho, D. Tarlow, R. van den Berg, Structured denoising diffusion models in discrete state-spaces, in: Advances in Neural Information Processing Systems 34 (NeurIPS 2021), 2021, arXiv:2107.03006.
URL `https://proceedings.neurips.cc/paper/2021/hash/958c530554f78bcd8e97125b70e6973d-Abstract.html`

[2] S. S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J. T. Chiu, A. Rush, V. Kuleshov, Simple and effective masked diffusion language models, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024), 2024, pp. 130136–130184, arXiv:2406.07524. doi:10.52202/079017-4135.

[3] J. Ou, S. Nie, K. Xue, F. Zhu, J. Sun, Z. Li, C. Li, Your absorbing discrete diffusion secretly models the conditional distributions of clean data, in: International Conference on Learning Representations (ICLR 2025), 2025, arXiv:2406.03736.
URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a365e37c18fb91af547a2f0012a89e98-Abstract-Conference.html

[4] A. Swerdlow, M. Prabhudesai, S. Gandhi, D. Pathak, K. Fragkiadaki, Unified multimodal discrete diffusion, arXiv:2503.20853v1 (preprint; no published version found in OpenAlex on 2026-10-28) (2025). arXiv:2503.20853.

[5] L. Yang, Y. Tian, B. Li, X. Zhang, K. Shen, Y. Tong, M. Wang, MMaDA: Multimodal large diffusion language models, in: Advances in Neural Information Processing Systems 38 (NeurIPS 2025), 2025, pp. 154021–154061, arXiv:2505.15809. doi:10.52202/085713-4636.

[6] Q. Shi, J. Bai, Z. Zhao, W. Chai, K. Yu, J. Wu, Y. Tong, X. Li, X. Li, S. Yan, Muddit: Liberating generation beyond text-to-image with a unified discrete diffusion model, arXiv:2505.23606v5. The arXiv comment states ICLR 2026; the proceedings record was not opened (2025). arXiv:2505.23606.

[7] S. Li, K. Kallidromitis, H. Bansal, A. Gokul, Y. Kato, K. Kozuka, J. Kuen, Z. Lin, K.-W. Chang, A. Grover, LaViDa: A large diffusion language model for multimodal understanding, in: Advances in Neural Information Processing Systems 38 (NeurIPS 2025), 2025, arXiv:2505.16839.
URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/975affbe7b5f3b55fba62247b6877b1c-Abstract-Conference.html

[8] Z. Chair, P. K. Varshney, Optimal data fusion in multiple sensor detection systems, IEEE Transactions on Aerospace and Electronic Systems AES-22 (1) (1986) 98–101. doi:10.1109/TAES.1986.310699.

[9] R. R. Tenney, N. R. Sandell, Detection with distributed sensors, IEEE Transactions on Aerospace and Electronic Systems AES-17 (4) (1981) 501–510, a short "Additions and Changes" note exists: IEEE Trans. Aerosp. Electron. Syst. AES-17(5):736, doi:10.1109/TAES.1981.309118. doi:10.1109/TAES.1981.309178.

[10] B. Khaleghi, A. Khamis, F. O. Karray, S. N. Razavi, Multisensor data fusion: A review of the state-of-the-art, Information Fusion 14 (1) (2013) 28–44, a corrigendum exists: Information Fusion 14(4):562, doi:10.1016/j.inffus.2012.10.004 (content not read). doi:10.1016/j.inffus.2011.08.001.

[11] A. Vempaty, L. Tong, P. K. Varshney, Distributed inference with Byzantine data: State-of-the-art review on data falsification attacks, IEEE Signal Processing Magazine 30 (5) (2013) 65–75. doi:10.1109/MSP.2013.2262116.

[12] Q. Zhang, Y. Wei, Z. Han, H. Fu, X. Peng, Q. Hu, C. Deng, C. Xu, J. Wen, D. Hu, C. Zhang, Multimodal fusion on low-quality data: A comprehensive survey, Information Fusion 135 (2026) 104437. doi:10.1016/j.inffus.2026.104437.

[13] Z. Shao, W. Dou, Y. Pan, Dual-level deep evidential fusion: Integrating multimodal information for enhanced reliable decision-making in deep learning, Information Fusion 103 (2024) 102113. doi:10.1016/j.inffus.2023.102113.

[14] L. Huang, S. Ruan, P. Decazes, T. Denœux, Deep evidential fusion with uncertainty quantification and reliability learning for multimodal medical image segmentation, Information Fusion 113 (2025) 102648. doi:10.1016/j.inffus.2024.102648.

[15] A. S. Willsky, A survey of design methods for failure detection in dynamic systems, Automatica 12 (6) (1976) 601–611. doi:10.1016/0005-1098(76)90041-8.

[16] Z. Gao, C. Cecati, S. X. Ding, A survey of fault diagnosis and fault-tolerant techniques—Part I: Fault diagnosis with model-based and signal-based approaches, IEEE Transactions on Industrial Electronics 62 (6) (2015) 3757–3767. doi:10.1109/TIE.2015.2417501.

[17] H. M. Finn, R. S. Johnson, Adaptive detection mode with threshold control as a function of spatially sampled clutter-level estimates, RCA Review 29 (3) (1968) 414–464, page range 414–464 per CiNii; OpenAlex gives 414–463. No DOI.

[18] H. Rohling, Radar CFAR thresholding in clutter and multiple target situations, IEEE Transactions on Aerospace and Electronic Systems AES-19 (4) (1983) 608–621. doi:10.1109/TAES.1983.309350.

[19] G. Wang, Y. Schiff, S. S. Sahoo, V. Kuleshov, Remasking discrete diffusion models with inference-time scaling, in: Advances in Neural Information Processing Systems 38 (NeurIPS 2025), 2025, pp. 163488–163545, arXiv:2503.00307. doi:10.52202/085713-4927.

[20] J. Kim, S. Kim, T. Lee, D. Z. Pan, H. Kim, S. Kakade, S. Chen, Fine-tuning masked diffusion for provable self-correction, arXiv:2510.01384v4 (preprint; no venue found in the arXiv record) (2025). arXiv:2510.01384.

[21] Y. Schiff, O. Belhasin, R. Uziel, G. Wang, M. Arriola, G. Turok, R. Zilberstein, M. Elad, V. Kuleshov, Learn from your mistakes: Self-correcting masked diffusion models, arXiv:2602.11590v4. The arXiv comment states NeurIPS 2026; the proceedings record was not opened (2026). arXiv:2602.11590.

[22] Z. Huang, Y. Wang, Z. Chen, G.-J. Qi, Don't settle too early: Self-reflective remasking for diffusion language models, arXiv:2509.23653v1 (preprint) (2025). arXiv:2509.23653.

[23] K. Zhai, S. Mollah, Z. Wang, M. Shah, CORE: Context-robust remasking for diffusion language models, arXiv:2602.04096v3 (preprint) (2026). arXiv:2602.04096.

[24] L. Yao, Remask, don't replace: Token-to-mask refinement in diffusion language models, arXiv:2604.18738v3 (preprint). Supersedes the withdrawn arXiv:2605.26436, which must not be cited (2026). arXiv:2604.18738.

[25] J. Lezama, H. Chang, L. Jiang, I. Essa, Improved masked image generation with Token-Critic, in: Computer Vision – ECCV 2022, Lecture Notes in Computer Science, 2022, pp. 70–86, arXiv:2209.04439. doi:10.1007/978-3-031-20050-2_5.

[26] Y. Ouyang, P. Hu, Z. Wan, Z. Wang, L. Xie, D. Bespalov, Y. N. Wu, G. Cheng, H. Zha, Q. Sun, Training-free self-correction for multimodal masked diffusion models, arXiv:2602.02927v1 (preprint). The name FreeCorrection appears only in the abstract's code URL (2026). arXiv:2602.02927.

[27] M.-Q. Le, A. Comas, A. Lattas, S. Moschoglou, P. Vélez, A. Raj, A. Germuth, T. Beeler, D. Samaras, D. Qiu, Concurrent image understanding and generation: Self-correcting coupled Markov jump processes, arXiv:2607.13188v1 (preprint). The abstract names the sampler CO2Jump (2026). arXiv:2607.13188.

[28] G. Jhingran, Trust but verify: Designing region-inhomogeneous forward dynamics for discrete diffusion, arXiv:2610.22380v1 (preprint). The arXiv comment states acceptance at the NeurIPS 2026 workshop "AI for Stochastic Dynamics" (2026). arXiv:2610.22380.

[29] G. Gomes, B. Martins, C. Zerva, A conformal risk control framework for granular word assessment and uncertainty calibration of CLIPScore quality estimates, in: Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 12348–12365, arXiv:2504.01225. doi:10.18653/v1/2025.findings-acl.638.

[30] G. Zhou, Y. Lu, Burden of proof: Calibrated evidential arbitration for asymmetric decision-level fusion in vision–language recognition, Information Fusion 139 (2027) 104860. doi:10.1016/j.inffus.2026.104860.

[31] A. Lou, C. Meng, S. Ermon, Discrete diffusion modeling by estimating the ratios of the data distribution, arXiv:2310.16834v3. The arXiv comment states ICML 2024 (oral); the proceedings record was not opened (2023). arXiv:2310.16834.

[32] S. Nie, F. Zhu, Z. You, X. Zhang, J. Ou, J. Hu, J. Zhou, Y. Lin, J.-R. Wen, C. Li, Large language diffusion models, in: Advances in Neural

Information Processing Systems 38 (NeurIPS 2025), 2025, pp. 56354–56392, arXiv:2502.09992. doi:10.52202/085713-1689.

[33] H. Chang, H. Zhang, L. Jiang, C. Liu, W. T. Freeman, MaskGIT: Masked generative image transformer, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 11305–11315, arXiv:2202.04200. doi:10.1109/CVPR52688.2022.01103.

[34] Y. Fang, Z. Yang, Z. Chen, Z. Zhao, J. Zhou, Enhancing vision-language model reliability with uncertainty-guided dropout decoding, in: Advances in Neural Information Processing Systems 38 (NeurIPS 2025), 2025, pp. 165747–165772, arXiv:2412.06474. doi:10.52202/085713-4991.

[35] G. Wan, Q. Li, G. Fang, X. Ma, X. Wang, Self-purification mitigates backdoors in multimodal diffusion language models, arXiv:2602.22246v1 (preprint). The abstract names the method DiSP (Diffusion Self-Purification) (2026). arXiv:2602.22246.

[36] J. Salazar, D. Liang, T. Q. Nguyen, K. Kirchhoff, Masked language model scoring, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 2699–2712, arXiv:1910.14659. doi:10.18653/v1/2020.acl-main.240.

[37] C. Kauf, A. Ivanova, A better way to do masked language model scoring, in: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 2023, pp. 925–935, arXiv:2305.10588. doi:10.18653/v1/2023.acl-short.80.

[38] P. Li, uChecker: Masked pretrained language models as unsupervised Chinese spelling checkers, arXiv:2209.07068v1. The arXiv comment states COLING 2022; no Crossref record or DOI was found (2022). arXiv:2209.07068.

[39] B. Krojer, E. Poole-Dayan, V. Voleti, C. Pal, S. Reddy, Are diffusion models vision-and-language reasoners?, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), 2023, pp. 8385–8405, arXiv:2305.16397. doi:10.52202/075280-0367.

[40] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, New York, 2005. doi:10.1007/b106715.

[41] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends in Machine Learning 16 (4) (2023) 494–591, arXiv:2107.07511 (arXiv title: A gentle introduction to conformal prediction and distribution-free uncertainty quantification). doi:10.1561/2200000101.

[42] S. Bates, E. Candès, L. Lei, Y. Romano, M. Sesia, Testing for outliers with conformal p-values, The Annals of Statistics 51 (1) (2023) 149–178, arXiv:2104.08279. doi:10.1214/22-AOS2244.

[43] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, in: International Conference on Learning Representations (ICLR 2024), 2024, arXiv:2208.02814.
URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/f3549ef9b5ff520a7e41ff3cc306ab2b-Abstract-Conference.html

[44] P. S. Diao, T. Alves, B. Poussot, S. Azarian, A review of radar detection fundamentals, IEEE Aerospace and Electronic Systems Magazine 39 (9) (2024) 4–24. doi:10.1109/MAES.2022.3177395.

[45] J. Kittler, M. Hatef, R. P. W. Duin, J. Matas, On combining classifiers, IEEE Transactions on Pattern Analysis and Machine Intelligence 20 (3) (1998) 226–239. doi:10.1109/34.667881.

[46] D. B. Rubin, Inference and missing data, Biometrika 63 (3) (1976) 581–592. doi:10.1093/biomet/63.3.581.

[47] B. Wang, W. Li, A. Bradlow, A. Watt, A. T. Chan, E. Bazuaye, Multi-stage multimodal fusion network with language models and uncertainty evaluation for early risk stratification in rheumatic and musculoskeletal diseases, Information Fusion 120 (2025) 103068. doi:10.1016/j.inffus.2025.103068.

[48] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick, Microsoft COCO: Common objects in context, in: Computer Vision – ECCV 2014, Lecture Notes in Computer Science, 2014, pp. 740–755, arXiv:1405.0312. doi:10.1007/978-3-319-10602-1_48.

[49] P. Sun, Y. Jiang, S. Chen, S. Zhang, B. Peng, P. Luo, Z. Yuan, Autoregressive model beats diffusion: Llama for scalable image generation,

arXiv:2406.06525v1 (preprint). Source of the VQ-16 image tokenizer named in the method specification (2024). arXiv:2406.06525.

[50] Q. McNemar, Note on the sampling error of the difference between correlated proportions or percentages, Psychometrika 12 (2) (1947) 153–157. doi:10.1007/BF02295996.

[51] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.
URL https://www.jstor.org/stable/4615733

[52] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413. doi:10.1093/biomet/26.4.404.

[53] L. Li, Z. Long, Y. Shen, H. Gao, H. Cao, X. Sun, C. Shan, R. He, C. Fu, Omni-diffusion: Unified multimodal understanding and generation with masked discrete diffusion, in: Proceedings of the 43rd International Conference on Machine Learning, Vol. 306 of Proceedings of Machine Learning Research, 2026, pp. 68853–68867, arXiv:2603.06577.
URL https://proceedings.mlr.press/v306/li26dv.html

[54] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, W. Brendel, Benchmarking robustness in object detection: Autonomous driving when winter is coming, arXiv:1907.07484v2; the imagecorruptions package, commit c959e659c9 (2019). arXiv:1907.07484.

[55] R. Shekhar, S. Pezzelle, Y. Klimovich, A. Herbelot, M. Nabi, E. Sangineto, R. Bernardi, FOIL it! Find one mismatch between image and language caption, in: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2017, pp. 255–265, arXiv:1705.01359. doi:10.18653/v1/P17-1024.

Table 1: Closest works on the four axes of Section 2. Entries restate what the cited papers report, at the level of detail read for this study; "–" means not applicable. AR: autoregressive. FAP: false-alarm proportion.

Work	Tokens treated as suspect	Score	Threshold	Use of flagged tokens
Trust but Verify [28]	Corrupted context tokens, text only	Probability of the observed token at its unmasked position (finetuned 170M model)	Hand-annealed, edit budget from an assumed rate; no guarantee	Re-noise to mask, refill
Gomes et al. [29]	Caption words given an image	Change of CLIPScore under masking	Conformal risk control, high-probability bound, foil-labelled calibration	Flag only
CO2Jump [27]	Tokens generated by the sampler	Self-confidence and a cross-modal confidence term	Per-step quota from the schedule	Remask, resample
FreeCorrection [26]	Tokens committed by the sampler	Likelihood of the committed token with it remasked	Lowest scores per step, schedule tuned on a validation set	Remask, resample
UniDisc, repair demo [4]	Random positions	–	–	Random mask, reunmask, best of n by model likelihood
Dropout decoding [34]	Visual tokens in an AR vision-language model	Divergence of a logit-lens distribution from the image mean	Hand-set drop probabilities	Drop in several masks, vote
This paper	Input image tokens of a given pair	Masked likelihood in own modality, other modality withheld, normalised by the position marginal	Conformal risk control on clean pairs, expected FAP at most α	Withhold (marginalise) or re-infill

Table 2: Notation. Symbols follow the preregistration documents and the analysis code. A letter with a second use appears in a separate row. Symbols used only in the protocol are defined there.

Symbol	Meaning
$x = (t, v)$	one pair of caption tokens t (n tokens) and image tokens v ($m = 1\,024$ tokens)
$\tilde{x} = (\tilde{t}, \tilde{v})$, v°	the observed, possibly corrupted pair; the tokens of the clean image
$Z_j = \mathbb{1}[\tilde{v}_j \neq v_j^\circ]$	corrupted-token label (used only for evaluation)
M, $\varnothing^a$	the mask token (M means "unknown"); all tokens of modality a masked
$a, b \in \{T, I\}$	scored modality and the other modality (text, image)
$p_\theta(x_j = \cdot \mid \cdot)$	categorical of the network at masked position j
$\Pi_K = \{\Gamma_1, \ldots, \Gamma_K\}$	random partition of the positions of modality a into K complementary groups; $k(j)$ is the group of j
A_j, B_j, C_j, D_j	four context levels of Eq. (1); $D_j[y]$ is the vector over codes y and $D_j = D_j[x_j]$
q_j, u_j	hypothesis-free intra-modal evidence $B_j - D_j$ and suspicion $u_j = -q_j$
α, $\hat{\tau}$, $\widehat{F}_{\text{cal}}$	target false-alarm level, calibrated threshold, empirical calibration risk (Eq. 5)
G, R	gated (flagged) set and reliable set, the complement of G
c, c^+, c^-	a candidate caption, the true caption and a wrong caption
$\Lambda_j^c[y]$	log-probability of image code y at position j with the image fully masked and caption c visible
Δ_j, $\bar{\Delta}_R$, Y_R	per-token paired evidence, its mean over R, and the decision $\mathbb{1}[\bar{\Delta}_R > 0]$ (Eq. 7)
d_R, z_R	size-normalised evidence ratio and evidence z-score over R (Eq. 8)
S_r, n_{draw}	number of re-infill steps and of independent re-infill draws
S, ρ, μ, σ^2, μ_c, σ_c^2	additive statistic $\sum_j I_j \Delta_j$; corrupted fraction; mean and variance of Δ_j on clean and on corrupted tokens (Eqs. 9–11)
ϕ_0, ϕ_1	probability that a clean, respectively a corrupted, token is flagged in the signal-to-noise analysis; $\hat{\phi}_0, \hat{\phi}_1$ their observed values (called α, β, α_{obs}, β_{obs} in the preregistration and the code)
γ_0, γ_1, γ_{01}	correlation of Δ_j and Δ_k, $j \neq k$, for two clean tokens, two corrupted tokens, and one of each (Proposition 2; theory only)

Table 3: Corruptions. Realistic corruptions are walked in order by the headroom rule (Section 4.4); the first rung that passes is the operating point. Sanity corruptions have one setting.

Corruption	Rungs	Definition
JPEG (realistic)	quality 10, then 5	JPEG compression of the 512-px pixels, then tokenisation
Gaussian noise (realistic)	$\sigma = 0.10$, then 0.20	noise on pixels scaled to $[0, 1]$, clipped and rounded to 8 bit, then tokenisation
Pixel splice (realistic)	128×128, then 192×192 px	a region from a donor COCO image copied to a uniformly random location
Patch (sanity)	one 8×8 block	a block of the token grid replaced by uniform random codes
Swap (sanity)	15 % of tokens	tokens replaced by uniform random codes

Table 4: Arms of the image-direction analysis. R is the set of image tokens kept in the sum of Eq. (7).

Arm	Kept set R	Role
none	all 1024 tokens of the corrupted image	reference
GATE-M	tokens not flagged by the frozen gate (Algorithm 4)	primary arm
geometry-matched random gate	complement of Algorithm 8 applied to GATE-M's mask, 3 seeds	mechanism control
scattered random gate	complement of equally many random tokens, 3 seeds	mechanism control
changed-code gate	tokens whose code did not change	upper reference for sanity corruptions only
plain-B top-k	complement of the $\lvert G\rvert$ tokens with the lowest B_j	gate without calibration, equal size
s_3 top-k	complement of the $\lvert G\rvert$ tokens with the largest $1 - \max_y p_\theta$	uncertainty-ranked gate, equal size
pixel denoise	none, on a denoised copy of the image (JPEG and noise)	defence comparator
GATE on clean images	as GATE-M on uncorrupted images	negative control

Table 5: Primary family P: GATE-M against no gate on the 500 test items, image-direction verification accuracy (higher is better). b: items right under GATE-M and wrong without a gate; c: the reverse; $n_d = b + c$. Net difference $= (b - c)/n$, GATE-M minus none, with paired percentile bootstrap intervals over items (10 000 resamples) at 90 % (which carries the negative label), 95 % and 98.33 % (the Holm level of the first step). p: exact two-sided McNemar test (Eq. (17)); p_{Holm}: Holm over the three corruptions. No estimate is bold: no interval excludes zero. Source: `runs/P2-GATE-M4/analysis.json`, keys `primary.per_corruption.{jpeg,noise,splice}` and `outcome.label`.

Corruption	n	Acc. none	Acc. GATE-M	b	c	Net diff.	p (p_{Holm})
JPEG, quality 10	500	0.838	0.836	0	1	−0.002	1.00 (1.00)
	intervals: 90 % [−0.006, 0.000]; 95 % [−0.006, 0.000]; 98.33 % [−0.008, 0.000]						
Noise, $\sigma = 0.10$	500	0.676	0.676	0	0	0.000	1.00 (1.00)
	intervals: 90 % [0.000, 0.000]; 95 % [0.000, 0.000]; 98.33 % [0.000, 0.000]						
Splice, 128 px	500	0.876	0.872	1	3	−0.004	0.625 (1.00)
	intervals: 90 % [−0.010, 0.002]; 95 % [−0.012, 0.004]; 98.33 % [−0.014, 0.004]						

Table 6: Headroom check on the 100 headroom items (no gate flag computed). n_v: items with all values; T_1, T_2: paired t statistics of clean minus corrupted and changed-code gate minus corrupted, on d_R; a rung passes if $n_v \geq 90$, $T_1 \geq 2$ and $T_2 \geq 2$, and the first passing rung is the operating point. D_c: items whose correctness differs between the changed-code gate and no gate. Changed codes: mean number of the 1 024 codes that differ from the clean tokenisation. Accuracy drop: clean minus corrupted accuracy without a gate. Rungs after a passing first rung were computed but not used. No interval is reported, so nothing is bold. $D^* = \max(19, 18, 2) = 19$ over the operating points gives rule R2 ($18 \leq D^* < 29$): 500 test items, endpoint Y. Source: `runs/P2-GATE-M3/family.json`, keys `rungs.*` and `n_rule.*`.

Rung	n_v	T_1	T_2	Passes	D_c	Changed codes	Accuracy drop	Operating point
JPEG, quality 10	99	6.03	3.30	yes	19	986.7	0.06	yes
JPEG, quality 5	95	5.16	1.54	no	26	1010.5	0.04	–
Noise, $\sigma = 0.10$	100	4.20	4.38	yes	18	949.4	0.14	yes
Noise, $\sigma = 0.20$	100	3.79	4.27	yes	19	998.4	0.22	–
Splice, 128 px	100	5.33	4.05	yes	2	237.2	0.01	yes
Splice, 192 px	100	7.93	7.34	yes	3	387.7	0.03	–

Table 7: Equal-size gates on the 500 test items: accuracy of GATE-M minus each comparison gate, and of the changed-code gate minus no gate, with paired percentile bootstrap 95 % intervals over items (10 000 resamples). Geometry-matched and scattered random gates are averaged over three seeds per item. Descriptive (exploratory); no p-value. Bold: the 95 % interval excludes zero. Source: `runs/P2-GATE-M4/analysis.json`, keys `mechanism.{jpeg10,noise0.1,splice128}.y.*`.

Corruption	GATE-M minus					Changed-code minus none
	geometry-matched	scattered	plain-B top-k	s_3 top-k	changed-code	
JPEG	0.000	−0.002	0.000	0.000	**0.082**	**−0.084**
	[−0.002, 0.002]	[−0.007, 0.001]	[−0.006, 0.006]	[−0.008, 0.008]	[0.044, 0.120]	[−0.122, −0.046]
Noise	0.001	0.000	−0.002	0.000	**−0.070**	**0.070**
	[0.000, 0.002]	[0.000, 0.000]	[−0.006, 0.000]	[0.000, 0.000]	[−0.104, −0.036]	[0.036, 0.104]
Splice	−0.001	−0.004	**−0.008**	−0.006	−0.008	0.004
	[−0.009, 0.007]	[−0.011, 0.002]	[−0.016, −0.002]	[−0.016, 0.002]	[−0.022, 0.006]	[−0.010, 0.018]

Table 8: Clean false-alarm check on the 500 clean test pairs. $\hat{F}$: mean over items of the flagged share of an item's 1 024 tokens; interval: percentile bootstrap 95 % over items (10 000 resamples). $\hat{F}_{\text{cal}}$: flagged share on the 300 calibration pairs at the fitted threshold. Z_{cal}: calibration statistic of Appendix C.1; a violation is flagged above 2.13. Nothing is bold (no interval of a difference). Source: `runs/P2-GATE-M4/analysis.json`, keys `calibration.fap_check.per_alpha.*`; `runs/P2-GATE-M2/threshold.json`, keys `thresholds.*.r_hat`.

α	$\hat{F}_{\text{cal}}$	$\hat{F}$ (test) [95 %]	Z_{cal}	Reading
0.01	0.0067	0.0064 [0.0061, 0.0067]	−16.12	no evidence of violation
0.05	0.0468	0.0456 [0.0442, 0.0470]	−3.59	no evidence of violation
0.10	0.0970	0.0956 [0.0928, 0.0985]	−1.76	no evidence of violation

Table 9: Flags of the gate at $\alpha = 0.05$ on the corrupted test images, token level, pooled over the 500 items. Changed share: share of tokens whose code differs from the clean tokenisation (ρ). Hit rate: share of changed tokens flagged. False-alarm proportion: share of unchanged tokens of corrupted items flagged (not covered by the guarantee). Precision: share of flagged tokens that changed. Point estimates; no interval was computed, so nothing is bold. Patch and swap are the sanity corruptions of the positive control. Source: `runs/P2-GATE-M4/analysis.json`, keys `calibration.per_condition.*.{precision,recall,fap_on_clean_tokens}` and `calibration.per_condition.*.prop4.rho`.

Corruption	Changed share	Hit rate	False-alarm proportion	Precision
JPEG, quality 10	0.961	0.061	0.037	0.976
Noise, $\sigma = 0.10$	0.928	0.033	0.036	0.922
Splice, 128 px	0.227	0.049	0.044	0.248
Patch (sanity)	0.062	0.750	0.050	0.498
Swap (sanity)	0.150	0.859	0.089	0.631

Table 10: Positive control PS (pipeline check only): GATE-M against no gate on the sanity corruptions, 500 test items, image-direction verification accuracy. Columns as in Table 5: paired percentile bootstrap intervals over items (10 000 resamples) at 95 % and 98.33 %; exact two-sided McNemar p; Holm over two. No estimate is bold: no interval excludes zero. Source: `runs/P2-GATE-M4/analysis.json`, keys `ps.per_corruption.{patch,swap}` and `ps.positive_control`.

Corruption	n	Acc. none	Acc. GATE-M	b	c	Net diff.	p (p_{Holm})
Patch, 8×8 codes	500	0.870	0.870	2	2	0.000	1.00 (1.00)
	intervals: 95 % [−0.008, 0.008]; 98.33 % [−0.010, 0.010]						
Swap, 15 % of codes	500	0.870	0.876	8	5	0.006	0.58 (1.00)
	intervals: 95 % [−0.008, 0.020]; 98.33 % [−0.012, 0.024]						

Table 11: E-dec on the accuracy scale, primary cells (descriptive; no test). T: clean minus corrupted accuracy without a gate; U: residual loss under the oracle (changed-code) gate; R: realised recovery by GATE-M; V: gap to the oracle gate; $T = U + R + V$. Paired percentile bootstrap 95 % intervals over items, items resampled jointly (10 000 resamples). n: complete items (two JPEG items have no finite changed-code value). Bold: the 95 % interval excludes zero. Source: `runs/P2-GATE-M4/edec.json`, keys `cells.{P1,P2,P3}.accuracy_scale.tau.*`.

Cell	n	T	U	R	V
JPEG (P1)	498	**0.044** [0.018, 0.070]	**0.124** [0.088, 0.161]	−0.002 [−0.006, 0.000]	**−0.078** [−0.116, −0.040]
Noise (P2)	500	**0.204** [0.166, 0.244]	**0.134** [0.098, 0.172]	0.000 [0.000, 0.000]	**0.070** [0.036, 0.106]
Splice (P3)	500	0.004 [−0.006, 0.014]	0.000 [−0.016, 0.016]	−0.004 [−0.012, 0.004]	0.008 [−0.006, 0.024]

Table 12: E-pred, primary cells (descriptive; family E withdrawn, no test). s_c: committed sign of the plug-in signal-to-noise change (Eq. (11)) from the fit chunk, with the share of fit resamples giving the same sign; o_c: sign of $b - c$ for GATE-M against none; o_c^*: the same sign only where the 95 % paired bootstrap interval of the net difference excludes zero. Net difference with its paired percentile bootstrap 95 % interval over items. Agreement $A/K_{\text{eval}} = 2/2$ (Clopper–Pearson 95 % interval 0.158 to 1.000); rival rules: "always +" 0/2, "− for primary cells, + for positive-control cells" 2/2. No estimate is bold: no interval excludes zero. Positive-control cells are reported in the text. Source: `runs/P2-GATE-M4/epred_score.json`, keys `cells.{P1,P2,P3}` and `P`.

Cell	predicted sign s_c	Same-sign share	b	c	Net diff. [95 %]	observed sign o_c	o_c^*	Agree
JPEG (P1)	–	1.000	0	1	−0.002 [−0.006, 0.000]	–	none	yes
Noise (P2)	–	1.000	0	0	0.000 [0.000, 0.000]	none	none	excluded ($b = c$)
Splice (P3)	–	1.000	1	3	−0.004 [−0.012, 0.004]	–	none	yes

Table 13: Exploratory (Tier 3): the hard gate GM (GATE-M) against detector-free combiners (MED: median; TM-G: trimmed mean removing $\lfloor g_i/2 \rfloor$ tokens from each tail, g_i the gate's flag count; TM20: 20 % trimmed mean), discounting of the gate-flagged tokens (SOFT-G) and reliability weighting of every token (SOFT-A), on the same Δ_j and items ($n = 500$ per cell; primary cells and the clean cell). Acc.: accuracy of the comparator. b/c: items right only under the comparator / only under GM. Δ: comparator minus GM with its percentile bootstrap 95 % interval (descriptive; 10 000 resamples, one index matrix per cell). Exact: Clopper–Pearson 95 % interval for b/n_d, which gives the reading. Bound: B (exact 95 % upper bound on the discordance rate) for $n_d < 10$, MDD for $n_d \geq 10$. Reading: above or below GM, not distinguishable (n.d.) or identical; †: counts as a finding against the added value of the detector (n.d. or identical for a detector-free combiner with bound at most 0.05; primary cells only). Bold: the exact interval excludes one half. 72 intervals in this comparison; no p-value and no multiplicity adjustment. Limits: (a) no sign-aware detector-free estimator is computed, because the direction in which a corruption moves Δ_j is defined by the true caption and such a combiner would use the label; (b) SOFT-G's weights lie token by token between GM's and no gate's, so it is not reliability weighting of every token; (c) when the gate's score is near chance the flagged count is close to the false-alarm count, so TM-G approximates no gate trimmed at about that count and TM-G minus GM largely repeats the primary contrast; (d) at this n many contrasts are uninformative, as the bound shows. Source: `runs/P2-GATE-M4/tier3_fusion.json`, keys `cells.*.arms`, `cells.*.contrasts.*_minus_GM`, `clean.arms`, `clean.contrasts`.

Cell	Comparator	Acc.	b/c	n_d	Δ [95 %, descriptive]	Exact 95 %	Bound	Reading
JPEG, quality 10	(GM 0.836; no gate 0.838; mean flagged tokens 61.8)							
	MED	0.804	4/20	24	**−0.032** [−0.052, −0.014]	0.047–0.374	MDD 0.027	below GM
	TM-G	0.834	2/3	5	−0.002 [−0.010, 0.006]	0.053–0.853	B 0.023	n.d.†
	TM20	0.816	2/12	14	**−0.020** [−0.034, −0.006]	0.018–0.428	MDD 0.021	below GM
	SOFT-G	0.838	1/0	1	0.002 [0.000, 0.006]	0.025–1.000	B 0.011	n.d.
	SOFT-A	0.838	8/7	15	0.002 [−0.014, 0.018]	0.266–0.787	MDD 0.022	n.d.
Noise, $\sigma = 0.10$	(GM 0.676; no gate 0.676; mean flagged tokens 33.9)							
	MED	0.678	10/9	19	0.002 [−0.016, 0.020]	0.289–0.756	MDD 0.025	n.d.†
	TM-G	0.676	1/1	2	0.000 [−0.006, 0.006]	0.013–0.987	B 0.014	n.d.†
	TM20	0.682	6/3	9	0.006 [−0.006, 0.018]	0.299–0.925	B 0.034	n.d.†
	SOFT-G	0.676	0/0	0	0.000	–	B 0.007	identical
	SOFT-A	0.676	9/9	18	0.000 [−0.016, 0.016]	0.260–0.740	MDD 0.024	n.d.
Splice, 128 px	(GM 0.872; no gate 0.876; mean flagged tokens 46.0)							
	MED	0.840	1/17	18	**−0.032** [−0.050, −0.016]	0.001–0.273	MDD 0.024	below GM
	TM-G	0.866	0/3	3	−0.006 [−0.014, 0.000]	0.000–0.708	B 0.017	n.d.†
	TM20	0.846	0/13	13	**−0.026** [−0.040, −0.014]	0.000–0.247	MDD 0.020	below GM
	SOFT-G	0.876	2/0	2	0.004 [0.000, 0.010]	0.158–1.000	B 0.014	n.d.
	SOFT-A	0.880	8/4	12	0.008 [−0.006, 0.022]	0.349–0.901	MDD 0.020	n.d.
Clean pairs, cost of each combiner	(GM 0.882; no gate 0.880; mean flagged tokens 46.7)							
	MED	0.852	1/16	17	**−0.030** [−0.046, −0.014]	0.001–0.287	MDD 0.023	below GM
	TM-G	0.874	1/5	6	−0.008 [−0.018, 0.000]	0.004–0.641	B 0.026	n.d.
	TM20	0.854	0/14	14	**−0.028** [−0.044, −0.014]	0.000–0.232	MDD 0.021	below GM
	SOFT-G	0.880	0/1	1	−0.002 [−0.006, 0.000]	0.000–0.975	B 0.011	n.d.
	SOFT-A	0.884	6/5	11	0.002 [−0.012, 0.014]	0.234–0.833	MDD 0.019	n.d.

Table 14: Exploratory (Tier 3), positive-control cells only (pipeline check; never pooled with or shown beside the primary cells). Columns, comparators, bound, reading and bold rule as in Table 13; $n = 500$ per cell; "against added value" is not read on these cells. Source: `runs/P2-GATE-M4/tier3_fusion.json`, keys `ps_cells.{patch,swap}.arms` and `ps_cells.{patch,swap}.contrasts.*_minus_GM`.

Cell	Comparator	Acc.	b/c	n_d	Δ [95 %, descriptive]	Exact 95 %	Bound	Reading
Patch	(GM 0.870; no gate 0.870; mean flagged tokens 96.5)							
	MED	0.846	2/14	16	**−0.024** [−0.040, −0.010]	0.016–0.383	MDD 0.022	below GM
	TM-G	0.858	2/8	10	−0.012 [−0.024, 0.000]	0.025–0.556	MDD 0.018	n.d.
	TM20	0.846	2/14	16	**−0.024** [−0.040, −0.010]	0.016–0.383	MDD 0.022	below GM
	SOFT-G	0.870	0/0	0	0.000	–	B 0.007	identical
	SOFT-A	0.880	8/3	11	0.010 [−0.002, 0.024]	0.390–0.940	MDD 0.018	n.d.
Swap	(GM 0.876; no gate 0.870; mean flagged tokens 209.7)							
	MED	0.838	1/20	21	**−0.038** [−0.056, −0.020]	0.001–0.238	MDD 0.025	below GM
	TM-G	0.852	1/13	14	**−0.024** [−0.040, −0.010]	0.002–0.339	MDD 0.021	below GM
	TM20	0.846	0/15	15	**−0.030** [−0.046, −0.016]	0.000–0.218	MDD 0.021	below GM
	SOFT-G	0.878	2/1	3	0.002 [−0.004, 0.010]	0.094–0.992	B 0.017	n.d.
	SOFT-A	0.892	10/2	12	**0.016** [0.004, 0.030]	0.516–0.979	MDD 0.019	above GM

Table 15: Status of every family and analysis of the main run (status only, no estimate). Nothing is bold. Sources: the run files cited in each subsection of Section 5.

Family or analysis	Status	Section
P (headline)	run, formal; $n = 500$ (rule R2); label negative	5.1
S-M	run; no hypothesis tested (gatekept: no primary rejection with $b > c$)	5.2
S-Z	run, formal	5.3
S-DN	run, formal	5.4
S-C	run, formal	5.5
PS	run, formal; positive control not met (audit triggered)	5.6
S-D, S-D2	run, formal; falsifier for the score met	5.7
E-dec, E-pred	run, descriptive	5.8, 5.9
S2	**[RESULT: status]**	5.10
B	**[RESULT: status]**	5.11
R	not run (gatekept: no primary rejection with $b > c$)	5.12
Suite	**[RESULT: status]**	5.13
Tier 3 comparison	run, exploratory (intervals only)	5.14
Coherence check	run, pipeline check; neither pre-stated reading	5.15

Table A.16: Pinned artefacts of the main run. "Pin file" is `runs/P2-GATE-CARD-V4/pins.json`.

Artefact	Identifier
Code, tests, guard (113 files)	commit `9c4e1e25cb0d1a77b78410e1613098d8c6ba1085` (final audited commit); moved by dated row A-1 to `f2f7b2e1 5e4f2b90545837d9e64207051c24a1cd`; per-file hashes in the pin file; manifest hash `65513677ba4d98adfa1a861db421520 0ad64b78aec1cceeaf0bcdca6b4664dec`
Analysis plan with Addenda A1 to A3b.12	`e33f078c45b92b94cbf658192e2fd815f8c6eafbffe53ba39b c0dd728982afb7` (LF)
Addendum A4	draft commit `40828e96`, revision r1 commit `f1671c91`
Fold file	LF `c19db5b3b9fabdeb89e8e686c58ae6c011f746529fbf9be 84f1c58c99c446786`; CRLF working copy `450edb543779de7 ee5e6a52422fd6507ef52f15ba3b956a8b05c2e561620f311`
Hard-negative pool, manifest, leak check	`cfd0f94facd0` (pool, LF), `070fb29a2ebc` (manifest, LF), `91f4b7968b54` (leak check, LF)
Frozen fit files of the descriptive family E-pred	`c89700dbc3ba` (fit), `5c535a3bda75` (signal-to-noise calculator), `8d7528bbb37b` (test file); audit record `26145a2a6b91`

Table A.17: Test families of the main run. n: planned number of items; m_{H}: hypotheses in the family's Holm correction. Each family has its own level 0.05; there is no control across families.

Family	n	Test	m_{H}	Not allowed
P (headline)	300, 500 or 300 with d_R	exact McNemar, GATE-M against none: JPEG, noise, splice; under rule R3 Wilcoxon on d_R	3	a claim from a rejection with $c > b$ in favour of the gate; "predicted", "explained", "caused by detection"
PS	as P	same test on patch and swap	2	"the gate works where detection is easy"; use as a demonstration
S2	300	exact McNemar I-GATE against I-RAND4 (kept draw), patch and splice; tested only if $n \geq 120$	2	"improves", "restores", gate detection as mechanism
B	300 pairs	exact McNemar fusion against better single direction; untestable at ceiling 0.95	1	a bound on foil noise; a statement about corruption
R	150	exact McNemar GATE-M against none per corruption of P^+; runs only if $\mathrm{P}^+ \neq \emptyset$	$\lvert\mathrm{P}^+\rvert$	"generalises"; pooling with UniDisc
S-M, S-Z, S-DN, S-D, S-D2, S-C	as P	mechanism, evidence ratio, denoising, detection, out-of-sample detection, clean control	3, 3, 2, 2, 2, 1	as in the protocol
E-dec	P fold	paired bootstrap intervals only	–	"treatment" or "detection" for its parts in abstract, highlights, conclusion
E-pred	P, PS cells	agreement of signs, Clopper–Pearson interval	–	"predicted", "explains", "supported"
Suite	100 + 100	eight cells, intervals only	–	any p-value in a claim; abstract or highlights

Table A.18: Crosswalk from preregistration and code names to the symbols of this paper.

Preregistration or code	This paper
`alpha`, `beta` in the signal-to-noise calculator	ϕ_0, ϕ_1
`alpha_obs`, `beta_obs`	$\hat{\phi}_0$, $\hat{\phi}_1$
CRC level α	α
partition group S_k	Γ_k
re-infill samples (`n_draw`, L)	n_{draw}
I-GATE, I-RAND4	GATE-I, random-gate best-of-four re-infill
Proposition 4 of the specification (plug-in signal-to-noise change)	Eq. (11)
Proposition 1 of the paper (conformal risk control applied)	Proposition 1

Table B.19: Evidence for the absence of a time input. Paths are relative to the UniDisc repository unless marked.

Claim	Evidence (file:line)
Time conditioning is off; the diffusion type is absorbing state, the parameterisation `subs`; no experiment overrides it	`configs/config.yaml:12,14,15`
The project composes the official demonstration experiments and loads the interleaved checkpoint	project files `runs/env-2026-10-14/scripts/unidisc_setup.py:17,30-35` and `exp001/unidisc_net.py:46-47`
The official inference experiment sets a null noise level; the authors' samplers pass `None`	`configs/experiments/large_scale_high_res_interleaved_inference.yaml:30-31`; `model_eval.py:2057,2086,3019,3060,3186`
The forward pass accepts a null noise level only when this is allowed	`model.py:660-662,691,729`
No timestep embedder and no time modulation in the final layer without time conditioning	`models/dit.py:1105,1186-1188,1377-1379` and `models/dit.py:1064-1090`
A stored scaling flag is never read	`configs/model/extra_large.yaml:8`; `models/dit.py:1290`
The checkpoint has no time-embedding or time-modulation keys; the string "sigma" does not occur in it	key names read from the checkpoint pickle, no tensor loaded
Raw logits are returned before the step that copies observed positions	`model.py:784-788`

Table C.20: Network passes per quantity. K: complementary groups; N: candidate captions; S_r, n_{draw}: re-infill steps and draws.

Quantity	Passes	Depends on, and cache
B over all image tokens	K	image only (caption masked); per image
C for image tokens (image fully masked)	1	caption only; per caption, shared across images and arms
D for image or caption tokens	1	layout only (caption length); per length
A over all image tokens	K	image and caption
Image-direction evidence, all arms	2	the two captions; arms differ only in gathered entries
Text-direction evidence per candidate and arm	1	image, gated set
Re-infill (GATE-I)	$S_r n_{\text{draw}}$	image, gated set; shared by all candidates